

ECO101 - Introduction to Microeconomics

Lecture Notes

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Chapter 1

Preamble

My aim in this course is to give you a basic understanding of how individual economic agents make decisions in the face of scarcity. Economic agents include you and your friends and the grocer who sells you vegetables and Apple Inc. that made your phone. We all face scarcity, which is not limited to money or raw material. Time is a scarce resource. Motivation is a scarce resource. Indeed, I always seem to be short on both time and motivation early in the morning. We all face scarcity. And in the face of these scarcities, we have to make the *best* decisions possible. Do I have time to shower in the morning? How about breakfast? Will I have to take a CNG autorickshaw instead of a bus? Should I just go back to sleep and continue living my best life? Decisions, decisions.

At the end of this semester, if I have done my job properly, you will understand exactly what type of scarcities and constraints different economic agents face in their decision-making processes and how they react to them. If that sounds simple enough, that is because it is. If it sounds complicating to you, my task this semester will be to show you otherwise.

1.1 Reading this Document

I aim to update this document on a weekly-basis to reflect our class lectures. Most of the text is in regular format. When I introduce a new term that I expect you to learn, it will be boldfaced for the first time that it appears. **This is an example of a sentence in boldface.** Examples and incidental comments are italicized. *This is an example of an italicized sentence.* Definitions are given in margin-notes.

this is a margin note

I have been requested often by students to add practice questions at the end of each section to help you practice before exams. I will not do that since I deem it unnecessary. There is a sufficient number of practice problems available at the end of each chapter of the assigned textbook¹. I encourage you to make full use of that. We will solve some practice exercises from the book in class. It is your responsibility to solve the rest.

1.2 Disclaimer

1. This document is intended to be used as a supplementary resource for students in my sections. The aim of this document is to complement my lectures, nothing more! Do not use this as a comprehensive guide to ECO101. People smarter than I am have spent a

¹*Economics* by Parkin, Powell, & Matthews

larger portions of their lives writing books about precisely what I am going to talk about here. Your assigned textbook is one such example. It is available in LMS moodle. Read it. Learn it.

If you however insist on following this document, expect to face a copious amount of errors, omissions, and eccentricities. I take responsibility for the eccentricities proudly and the errors and omissions grudgingly.

2. The diagrams used in this document are lifted directly from the assigned textbook for this course. I have not sought the authors' or the publisher's permission to do so but since I am not using this document for any commercial purpose, I hope they will not mind.

Chapter 2

Introduction

2.1 What is this thing called Economics?

What is economics? What do you hope to learn from this course, Introduction to Microeconomics? Different textbooks define economics in different ways. Similarly, different instructors of ECO101 will define this economics thing to you very differently. That is alright.¹ All we need for this course is to get a working-definition of economics off the ground that allows us to undertake some preliminary analyses. This definition does not have to be all-encompassing and perfect. It just has to allow us to begin our enquiry.

So that brings us back to our original question: what is economics? My approach is to view economics as a tool.² A tool is any device that helps us perform a specific task or a function. But if economics is just a tool, that raises yet another question: What problem(s) does economics help us solve?

A hammer is a tool and the problems that it helps us solve are very easily identifiable inasmuch as when faced with a host of problems, it is easy to identify which of the problems can be solved by using a hammer and which problems cannot be solved with a hammer.

What problem(s) may we hope to solve with economics?

The problem we hope to solve is the one of trying to find the most efficient allocation and use of the limited resources that we have to satisfy our unlimited wants. A pauper may dream of a blue-collar minimum-wage job but the minimum-wage worker dreams of a supervisor's job, who in turn dreams of a white-collar job. A data-operator will like to have a job as a software engineer someday but the software engineer would one day like to be the next entrepreneur/innovator much like Bill Gates or Mark Zuckerberg. And Bill Gates, what does he dream of? Is he satisfied with what he has and just sit in his room all day laughing hysterically about the mountain of wealth at his disposal or does he hope to achieve more (not necessarily money) during what remains of his life? Is Mark Zuckerberg happy with Facebook and Instagram or does he dream of becoming our lizard-overlord someday? Whatever our personal situations may be, we always want the next best thing. Our wants are never quenched.

¹As you will notice during our lectures and exams, I do not care about the exact wordings of a definition.

²The same way *science* is just a tool. Science is not physics or chemistry or biology. Science, in its purest form, is merely a collection of rules to follow in our pursuit of an answer.

A person with one shirt wants two shirts but a person with two shirts wants three shirts. A person with ten shirts may think, “It’s time I went to a bespoke and got myself a nice blazer.” This is the concept of unlimited wants.

However, we do not always have the resources to satisfy all our wants. We may not have the money to buy one more shirt. We may be so busy that we do not have the time to go to a shop or to a tailor and get a new shirt made. There may not be any space left in our closet to fit one more shirt. The economy may not have access to the raw materials needed to make the fabric (cotton, for example) needed to make a new shirt. These are all examples of our unlimited wants being thwarted by limited resources (money, raw material, time, and any other constraint).

That is where economics comes in. The study of economics is primarily concerned with the efficient allocation of **limited resources** to satisfy our **unlimited wants**. Since we have unlimited wants and limited resources (scarcity), we have to make choices. When we make choices, we are choosing one bundle of goods and services in lieu of another – there is a **trade-off** involved. If we have multiple options to choose from and select one bundle in favor of all other, it is assumed that we have selected the option that we like best.³ The second-best option that was available to us, the best option sacrificed, is our **opportunity cost**.

If I am used to waking up at 8am in the morning but decide to sleep until 9am during a vacation day, there is a trade-off involved. I have decided to sacrifice one hour of being productive for one additional hour of sleep. When I decided to sleep between the hour of 8am and 9am, it was because this activity gave me the most benefit/pleasure/joy out of all other activities that I could have done by waking up – listening to some podcasts, reading a book, spending time with my family, going for a run, et cetera.

Out of all these activities forgone, the one that would have given me the most joy – reading a book for me – is my opportunity cost.

So, to reiterate: since we have scarcity, we have to make choices. When we make choices, there is a trade-off involved. And when we trade-off, we create opportunity cost.

Now let us talk about a slightly trickier concept that may not be intuitively obvious to everyone – more we engage in one activity, higher its opportunity. Or, in other words, the opportunity cost of every additional unit of an activity performed is higher than the previous unit. Effectively, what we mean to say is that (rational) people prefers a variety of activities to perform than just one.

Suppose you are used to taking 4 courses per semester but for the upcoming semester, you decide to take 5 courses. Your opportunity cost for the fifth course will be the time that you will spend attending lectures and studying for the fifth course.⁴ Suppose, for the semester after next, you decide to take 6 courses. Similarly, the opportunity cost for the sixth course will be the time that you will spend attending lectures and studying for the sixth course, but in addition, you may now be sleep-deprived because of the extra load and sleep deprivation is also a part of your opportunity cost now.

³since we are *rational* consumers

⁴Let us ignore financial cost for now and assume that your parents are paying for your educations and not you, although that may not be true for some of you.

What if you decide to take 7 courses? Time spent attending lectures and studying are still part of your opportunity cost, as is sleep deprivation, but now you may have other additional opportunity costs involved: finding time to maintain relationships with your friends and maintaining your mental wellbeing. So you see, for each additional course-load taken, your opportunity cost increases. The more you do of any one activity, the higher its opportunity cost.

2.2 Production Possibility Frontiers

A country can produce any number of goods (and services), using any number of resources. For example, within the borders of Bangladesh, quite likely, thousands of goods and services are produced every day, using thousands of different types of resources. But let us simplify things a little bit and assume that Bangladesh only produces two goods. Let us also assume that the two goods that Bangladesh produces are Pepsi and Lay's chips. Every year, Bangladesh will use all its resources behind the productions of Pepsi and Lay's chips and the people of the country consume only these two goods. I am sure we all have someone in our friend-circle who would quite enjoy that life.

We take a simplified 2-goods case for two reasons: analyzing a country that produces 2 goods and a country that produces 3 goods (or any other number of goods), is similar. If we can understand the dynamics behind a 2-goods case, we can easily understand the dynamics behind any other n -goods case.

Furthermore, graphically, it is easy to study the 2-goods case. A 3-goods case is very difficult to study graphically and a 4 or higher goods case, impossible.

Let me now ask the first important question: what portion of our resources should be spent behind the production of each good? That's easy, right? If the people of Bangladesh like Pepsi more than Lay's chips, more resources should be used in the production of Pepsi. If the reverse was true, more resources should be used in the production of Lay's chips. (In the extreme cases, we may spend all our resources producing only Pepsi, or we may spend all our resources producing only Lay's chips.) Usually, a country will choose to produce some combination of the two goods, i.e., split its resources in production of Pepsi and Lay's chips.⁵ So suppose we are at one such point, where we are using some part of our resources in producing Pepsi and the remaining resources in producing Lay's chips.

If we are producing a bundle of goods by using all our resources, and we want to increase the production of one good from that bundle, it should be obvious that we can only do that by decreasing the production of at least one other good – there is a tradeoff involved.

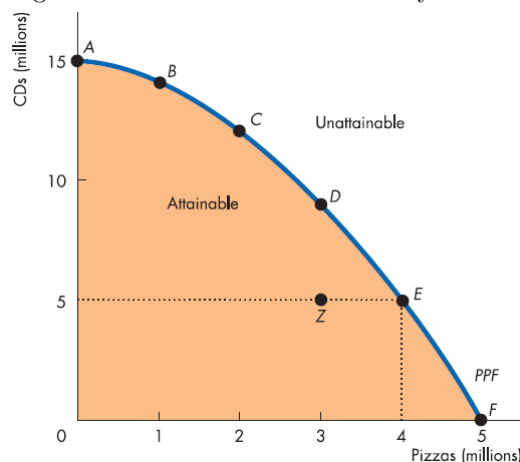
If we put the productions of Pepsi and Lay's chips on the x-axis and y-axis of a Cartesian-coordinate system respectively, we will get a concave curve that will show us the different combination amounts of the two goods that Bangladesh could produce, within a given period of time, by using all its resources. This curve is known as a **Production Possibility Frontier** (PPF), or, a Production Possibility Curve (PPC).

Any point on the PPF indicates that production is taking place efficiently, with full and proper usage of resources. Any movement along a PPF leads to a trade-off between the productions of two goods. Any point inside a PPF indicates inefficient use/allocation of resources. Any point outside a PPF is currently infeasible due to resource and technology constraints. **Technological**

⁵Can you think of a case when, given the option of producing two goods (any two goods, not necessarily Pepsi and Lay's chips), a country decides to produce only one?

A Production Possibility Frontier is a locus of points that show the different combinations of good that can be produced in a period of time, using the available technology and resources.

Figure 2.1: A Production Possibility Frontier



progress⁶ shifts the boundaries of a PPF outside, vice versa.

Refer to Figure 2.1. Suppose we can produce two goods, CDs and pizzas and our initial PPF is given by the blue-arc. That means that if we use all our resources to only produce CDs and no pizza, we can produce a maximum of 15 million CDs. Conversely, if we want to only produce pizza, we can produce a maximum of 5 million pizzas.

Usually, we will want to produce a combination of both CDs and pizzas. Every point on the PPF gives us a feasible combination of CDs and pizzas that we can produce using all our resources, for example, points A, B, C, D, E, or F. Each of these points represent full employment of available resources.

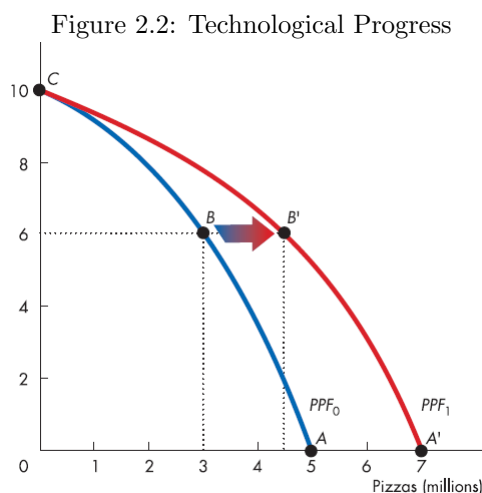
Any point inside the PPF, such as point Z, represents inefficient or under use of resources. resources are sitting idle and it is possible for us to produce more. All points outside the PPF is unfeasible at the moment. We do not have the necessary resources to produce them. However, technological progress allows us to produce more with the available resources. In such situations, the PPF shifts out.

Refer to figure 2.2. Improvements in the technology to makes pizzas now allow us to make more pizza using the same level of resources as before. Is it ever possible for the PPF to contract and move inside?

The bulging-out (concave to-the-origin) shape of PPF illustrates rising opportunity cost –the opportunity cost of X and Y increases as more is produced. To see this relationship, look at the moves from producing bundle A to bundle B, and from bundle C to bundle D. At bundle A, we are producing very few units of Y and if we want to have a little more of Y by moving to bundle B, we have to sacrifice a little amount of X. However, at bundle C, we are already producing a lot of units of Y. If we want even more Y by moving to bundle D, we have to sacrifice a lot of X.

The bulging-out (concave to the origin) shape of a PPF represents **increasing opportunity cost** of trade-off: the more we produce of good A, the more expensive it becomes to produce A, in terms of how much we have to sacrifice production of good B. As a result, we have to sacrifice an increasing number of good B to produce more of good A. This is because resources most appropriate at producing good A are used up first. As we continue to produce more of good A, less appropriate resources have to be used, requiring higher inputs for the same output of A.

⁶capital enhancement or human capital improvement



In ECO101, often, it may seem like I am simplifying things far too much and you may worry that the theories we work with in this course may not apply to a more practical and complex setup. For example, the assumption that only two goods are produced in Bangladesh, pizzas and CDs, may seem far too simplistic to some of you, if not downright absurd! However, how different do you think our analyses would have been if we had assumed ten goods instead of two? What if we had stayed with the two-good model, but instead of pizzas and CDs, put ‘manufactured goods’ and agricultural goods’ on the two axes?

Suddenly, the situation is not so absurd anymore!

Recall what we had talked about in the previous lecture. We have limited resources, which forces us to make choices, which leads to trade-offs, creating opportunity costs, and that opportunity costs increase as we partake more in one good or activity. Go back to the diagram of PPF and try to identify each of these concepts in the diagram. You will notice that our entire first lecture is captured within a simple PPF. Now try to link each of these concepts to the real-life situation of the Bangladesh economy. You start to notice that this simple concave curve, drawn on a simple Cartesian-coordinate system, is more versatile than may have first appeared and allows us to undertake plenty of useful analyses.

2.3 Absolute and Comparative Advantage

It is important to realize that production possibilities will differ from person to person since individuals have different abilities, skills-sets, resources at hand, et cetera. Given two goods, X and Y, and two people, A and B, in **autarky**, both A and B would be producing both goods, X and Y, for their individual consumption. However, in our everyday life, we see people trading with each other all the time. In fact, trade is the very basis of our modern economy. Why do we do that? Consider the following example.

Suppose person A can produce one unit of good A or one unit of good B in 3 minutes. Therefore, in an hour, he can produce 20 units of good X, 20 units of good Y, or a combination of both good X and good Y. Person B, on the other hand, can produce one unit of good X much quicker, in only 1.5 minutes. However, he is not very good in producing good Y, which takes him 6 minutes to make. Therefore, in an hour, he can make 40 units of good X, 10 units of good Y, or a combination of both good X and good Y. We have three combinations from the PPFs of both Person A and Person B.

Person A		Person B	
Good X	Good Y	Good X	Good Y
20	0	40	0
10	10	20	5
0	20	0	10

Draw the PPFs and mark the three points.

Person A has **absolute advantage** in the production of good Y, and person B has absolute advantage in the production of good X. Therefore, person A should concentrate on focusing good Y and person B should focus on producing good B.

Suppose they do and then they trade. Person A produce 20 units of good Y and trade 10 units of it for 20 out of the 40 units of good X that person B has produced. After trade, both person A and person B consumes 20 units of good X and 10 units of good Y. Plot these points of the PPFs. They are outside the PPFs! But how? Through trade. By focusing on the good we have an absolute advantage in producing, and then trading, we are able to consume at a point outside our PPF - a point we could not have reached on our own. What if a person has absolute advantage in producing both goods?

India		Bangladesh	
Cars	Clothes	Cars	Clothes
50	0	10	0
25	50	5	20
0	100	0	40

In this case, India has the absolute advantage in production of both Cars and Clothes. But we know that India and Bangladesh still trade. In fact, in 2018, Bangladesh and India traded goods worth over US\$10 billion. How can we explain such strong bilateral relationships between the two countries when India can produce more of all goods than Bangladesh?

To make sense of that, we need to introduce a new term called **comparative advantage**. Recall our discussion on opportunity cost. A country is said to have a comparative advantage in the production of a good if their opportunity cost lower. Consider the following opportunity cost table.

India		Bangladesh	
Cars	Clothes	Cars	Clothes
2	0.5	4	0.25

To produce one car, India has to sacrifice two articles of clothes but Bangladesh has to sacrifice four articles of clothes. Therefore, in comparative terms, making cars is cheaper in India than in Bangladesh. Similarly, to make one article of cloth, India has to sacrifice half a car but Bangladesh only has to sacrifice quarter of a car. Therefore, clothes can be made in Bangladesh cheaper than in India.

India should make 50 cars, Bangladesh should make 40 articles of clothing, and the two country should trade.

Remember: absolute advantage is concerned with production, comparative advantage is concerned with opportunity cost.

Here is the takeaway from this section: even though a person (or a firm's or a country's or any other economic agent's) production capacity is constrained by their PPFs, by trading with another entity with absolute or comparative advantage allows both parties to consume at a point beyond the individual PPFs. Verify this by drawing the PPFs from our examples and then finding the consumption point after trade.

This is known as **Gain from Trade**. This effectively, is the reason why people/firms/countries trade with each other. By trading, we are able to consume at a point beyond our productive capacities.

2.4 Basic Graphing Skills

Now let us shift our focus towards some graphing skills. From our next lecture, we will start to talk about Demand and Supply and those discussions involve a lot of graphs. If that worries you, know that this course does not have any higher mathematics requirements. What you need to know, you have already learned in your O Levels or SSC exams. I am just going to review it very quickly.

The equation of a straight-line is $y = mx + c$, where m and c are two real-numbers. m gives us the slope (rate of change) of the line, inasmuch as it tells us the rate of change of y due to a change in x . Therefore, if $m = 3$, then when the value of x changes by 1 – *unit*, the value of y changes by 3 units. Similarly, if $m = -2$, then every time x increases (decreases) by 1 – *unit*, y decreases (increases) by 2 *units*. Try to work out the different shapes a straight-line takes on due to different values of m . c gives us the vertical-intercept of the straight-line, or in other words, c is the value of y at which the line cuts the y – *axis*. Try to work out the different shapes that a straight-line takes on due to different values of c .

You all know that the slope of a straight-line is constant. But that is not always the case. Slopes can change, depending on the value of x . Relationships between x and y can be one of the following:

Types of Relationships	Slope
1 Increasing at an increasing rate	Positive, increases as x increases
2 Constant increasing	Positive, constant
3 Increasing at a decreasing rate	Positive, decreases as x increases
4 Decreasing at an increasing rate	Negative, decreases as x increases
5 Constant decreasing	Negative, constant
6 Decreasing at a decreasing rate	Negative, increases as x increases
7 Relationship with a maximum point	Relationship type 3 + type 4
8 Relationship with a minimum point	Relationship type 6 + type 1
9 Combinations	

The worst thing you can do is try to memorize these relationships. I can tell you right now that doing so will not help you in this course, or any other course you take. Try to understand what is going on. The easiest way to do that is by doing these two things: draw lines or curves that represent each of these 9 cases, and then try to come up with examples of such relationships between two phenomena from your life.

For example, a good example of relationship number 3 would put ‘hours studied’ on the x -axis and ‘knowledge/information acquired’ on the y -axis. Don’t you agree?

Chapter 3

Demand, Supply, and Market Equilibrium

3.1 Introduction to Demand

I went grocery shopping a few days ago. As I entered the store, I knew what I wanted to buy, and I had an idea about how much money I would spend. But in the store, I noticed that the prices of some of the good I wanted to buy had gone up since my previous visit to the store. I did not want to spend more money. Here is how I adjusted to the situation.

Since the price of the brand of tooth-paste I use, Pepsodent, had gone up, I bought tooth-paste of a different brand, Close-Up, which was cheaper. Instant coffee was more expensive as well, but I like the brand that I use, Maxwell House. Instead of buying another brand, I decided to buy a smaller packet than usual. I also wanted to buy a backpack. I no longer wanted to carry my old backpack that hung from both my shoulders. My taste had changed. I bought myself a new backpack with a sling-strap that would hang from only one shoulder.

Before leaving the store, it occurred to me that although I have my coffee black, everyone else at home prefers to have their coffee with milk and sugar. So I bought some milk and sugar as well. Since I had purchased a smaller packet of coffee, I decided to buy smaller packets of milk and sugar as well, even though their prices had not gone up. Since I was expecting a big bonus from work next month, I knew that I would be able to buy more coffee next month. But not the same brand, I decided as I left the store. The bonus would allow me to buy fewer packets of instant coffee and more packets of ground coffee, which is a bit more expensive, but tastes much better.

Do you think I was being overtly capricious in how I went about shopping? I bought less of coffee than usual since its price had gone up; but I also bought less of milk and sugar even though their prices had not changed. I had a perfectly good dual-strap backpack, but I decided to buy a new bag anyway. Then, I decided to buy more of ground coffee when I had more money. But I also decided to buy less of instant coffee when I would have more money. What is going on here? What is wrong with me? Why do my demands for different goods (and services, for example,

a haircut) rise and fall so impulsively? Can I ever be trusted with making decisions again after such behavior?

To answer that, we first need to answer this all important question: What does it mean to demand something?

To most of us, that is a very straight-forward question, easily answered. To demand something is to want it, is it not? I am thirsty and I want a bottle of water, so I have demand for a bottle of water. What else is there to it? Well, suppose while walking through a shopping mall, I come across a beautiful, state-of-the-art, 50-inch flat-screen, 4K TV and the first thing I say to myself is, “I want that.” Do I then have demand for that TV? Almost everyone who comes across that TV will probably want it, so does that mean that the demand for a 50-inch flat-screen 4K TV is around 7 billion, which is the population of the world?

Or how about this: suppose I love chocolates and out of all the different varieties of chocolates available in a store in Dhaka, my favorite chocolate is Ferrero Rochers. Every time I see Ferrero Rochers, I want to have it. But, suppose, I am also on a diet and I have decided that I will have no more sweets ever again. Do I still have demand for Ferrero Rochers? To effectively demand a good (or a service), we need to satisfy these three criteria:

- Want it
- Able to afford it
- Plan to buy it

I do not have demand for the 50-inch TV even though I want it because I cannot afford it. I do not have demand for Ferrero Rochers even though I want it and can afford it because I do not have any plans to buy it. However, that bottle of water I mentioned? Unless I am fasting and have enough money in my wallet, I can say that I have effective demand for that bottle of water.

So now we know what it means to have demand for a good or a service in the market. That brings us to the **Law of Demand** which states that a price change of a good or a service is inversely related with a change in the quantity demanded of that good or service.

I usually drink 5 bottles of Coke every week, buying each bottle for taka 20 each. However, if the price goes up to taka 25 each, maybe I will decide to have 4 bottles a week, instead of 5. And if the price was to fall down to taka 18 each, I would definitely have more than 5 bottles a week.

A rise in the price of a good or a service will, ceteris paribus, lead to a fall in the quantity demanded of that good or service, vice versa.

Let us think a little more about what this says. There was an existing price level in the market based on which, I was buying and consuming a certain quantity of a good (or a service). When that price level changed, I responded to the change by adjusting the level of my consumption. We are all used to such events. However, is price level the only thing that determines how much we buy and consume of a product, or are there other factors affecting our buying decisions?

3.1.1 Factors that Affect Demand

Beyond price level, what other factors affect our buying decisions? There can be many factors but we have listed the most common and important factors underneath:

1. Income level or size of the budget

When our income level goes up, our demand for certain goods and services go up because

*Income rise affects
the demand of
normal goods
positively and
inferior goods
adversely, vice
versa.*

we have more money at hand. These goods and services are **normal**. However, a rise in income can also lead to a fall in demand. In that case, the good or service is **inferior**.

A good example for this is the food-intake composition of people in Bangladesh. The diet for most people in this country consists mostly of rice and some vegetable. But as we move to wealthier households, we notice a rising consumption level of meat and fish and a falling consumption of vegetables. Therefore, we can say that vegetables are inferior goods and meat and fish are normal goods.

When I was a university student, much as you are right now, I had a very limited income. I used to tutor a few students and had to cover all my expenses from that. As you can guess, I had to be careful with how I spent my money. Therefore, I used to ride busses for my daily commute from my home in Dhanmondi to my University in Bashundhara in the morning and then back again in the evening. I did this for almost four years. However, once I graduated and got a job in Gulshan, my money situation improved and I started taking CNG auto-rickshaws. I had more money, so I could afford it.

*In this case, a CNG auto-rickshaw ride was a **normal good** for me. A rise in the income level leads to a rise in the demand of this good. Notice that the price level of using CNG auto-rickshaws (the fare) had not changed – it remained at the same. But for whatever that level may have been, my demand for it had gone up. This is called a rise in demand. Notice, however, that my demand for bus rides had gone down! Because I had higher income, my demand for that service was lower, despite there being no changes in its price level (the price of a ticket). Therefore, in this case, a bus ride was an **inferior good** for me.*

This is not to say that bus rides will always be an inferior good and CNG rides will always be a normal good. As soon as I got a few promotions and my income increased more, I stopped commuting with CNG auto-rickshaws and started taking an Uber. Due to a rise in my income, CNG rides became inferior. And if my income goes up even more, and I can afford to buy my own car, then Uber rides will become inferior as well.

And if someday I have my own private jet, even public airlines will be an inferior good to me. We can all dream, right?

2. Price of related goods

If the price of Coke goes up, customers will quickly reduce their demand of Coke and increase their demand for Pepsi. Coke and Pepsi are **substitute goods**. One can easily be substituted for another. On the contrary, if price of shoes goes up, the demand for socks will go down. Shoes and socks are **complementary goods**. They are used/consumed together and therefore, if there is a rise in price of one good (leading to a fall in quantity demanded of it), there will be an immediate fall in the demand of the other good.

Just how close a substitute/complement are the two goods we are interested in? For example, Coke and Pepsi are closer substitutes than tea and coffee. Pencils and sharpeners are closer complements than pencils and erasers.

*Can we come up with examples of **perfect substitute goods** and **perfect complement goods**?*

*A rise in the price
of a good affects
increase the demand
of its substitute and
decreases the
demand of its
complement, vice
versa.*

Coke and Pepsi are substitute goods, but not perfect substitutes. Most people may be willing to substitute one for another, but not everyone. There is brand-loyalty involved and some people only like the taste of one and not the other. But how about two street-vendors, selling vegetables from their carts, on the same street-corner? They are selling the same product with no differentiable qualities and if one of them tries to charge a higher price than the other vendor, he will lose all his customers. In this case, these generic, non-branded vegetables being sold by two vendors are perfect substitute goods.

Similarly, shoes and socks are very close complement goods but not nearly perfect. After all, some people wear shoes without socks and some people wear socks with their sandals (as weird as that may be). What is a good example of perfect complements? How about your left-shoe and your right-shoe? If the price of your right-shoe goes up, law of demand says that you will buy fewer right-shoes. But will you continue buying the same quantity of left-shoes? Unlikely.

Try to come up with 3-5 example of substitute and complement goods. Then try to come up with at least 1 new example of a pair of perfect substitute and perfect complement goods. It should not be very difficult.

3. Population

This is pretty obvious. More people, more demand. Population growth of course is a much slower, gradual change, since population grows very slowly, maybe at around 1% or 2% per year. But there are special circumstances. Refugee inflow for example leads to a sudden rise in population of a country and refugee outflow has the opposite effect. Outbreak of an infectious disease can very quickly decimate the population of a region (in 14th century Europe, the Black Plague killed off as much as 200 million people in a 7-year period).

The influx of Rohingya refugees in the south-eastern part of Bangladesh has led to a sharp increase in the population of that region. Now a good argument here may be that the Rohingyas have not come in to the country with any money and as such have not increased the demand of goods and services in the region. If this point occurred to you, very good. You are already starting to think like an economist.

It is true that Rohingyas are not engaging in buying and selling at the market place. However, do not forget the millions of dollars in aid money that has flown into the region from all over the world to help the Rohingyas. That has led to a sharp hike in inflow of aid workers, development field works, and other international consultants and agents from all over the world to move to Cox's Bazaar and the surrounding areas. An entire subsidiary industry has developed there, ensuring that the aid money is spent properly and benefits the right people. And that can be equated as a sharp rise in population of the region, leading to a sharp rise in demand.

Don't believe me? How about the fact that few miles away from the refugee camp, which is one of the poorest regions of the country, there is a North End coffee shop?

4. Tastes and preference

The clothes you liked last month may not be what you like anymore. The songs your

grandparents used to listen to are not the same songs you listen to. Tastes and preferences change over time. This change can be quick (from week to week or month to month) because an influencer told you so, or a more gradual process (over decades) because economic and social realities are different. For example, growing up, I never saw anyone enjoying Japanese cuisine in Dhaka. Few places that served Japanese food were not popular. But look around now. There are multiple stores selling Japanese food (and also authentic Chinese, Korean, and other East Asian delicacies) in almost every single popular streets of Dhaka. What happened? Did Japanese food suddenly become cheaper, leading to a rise in the quantity that is demanded?¹ Or did people's taste change over a number of years and that led to a gradual rise in the demand of Japanese food?

5. Seasonality

Demand for ice-cream is not the same around the year. I have a higher demand for ice-cream during the summer months than during the winter months. During winter, however, I have a higher demand for cardigans. During the rainy months, I have a high demand for umbrellas. And for a one-month period once every 4 years, the demands for replica flags of Brazil and Argentina are sky-high in Bangladesh, but almost non-existent at other times. Some goods have seasonal demands –their demands go up and down according to some easily predictable factors.

6. Expected future price

If I expect the price of a good to go up in the future, I will want to buy more of it now. For example, if I expect the government to impose a new, higher tax on import of cars in the next Annual Budget, if I want to buy a new car, I will not wait until next year – I will buy the car immediately, before price goes up. If I expect the opposite, that the government will remove all existing taxes on import of cars, even if I have the money at hand right now, I may wait until next year to buy a car, when prices are lower. Can you think of some other examples?

7. Saturation level²

How often have you discovered a new song, listened to it on repeat for hours, and ended up hating the very sound of that song? Or, have you ever visited your favorite restaurant so often that you started disliking the taste of your favorite meal? These are all examples of over-saturation with a product leading to a fall in its demand. The reverse is also true: if you have not been able to visit your favorite restaurant for three-months, next time you go there, instead of getting your regular order, you may get two of each. In this case, "under"-saturation, if that is a thing, has led to a rise in demand.

3.1.2 Demand and Quantity Demanded

The Law of Demand told us that whenever the existing price-level in the market changes, we react to the change by adjusting how much of that good we buy – our quantity demanded

¹I wish that was the case, but it is quite the opposite. Food prices in Bangladesh never go down. In fact, the cost of eating out in Dhaka is, as you all know, exorbitantly high. You can buy a Japanese bento-box in Vancouver, one of the most expensive cities to live in in the world, for half the price of what you would have to pay in Dhaka.

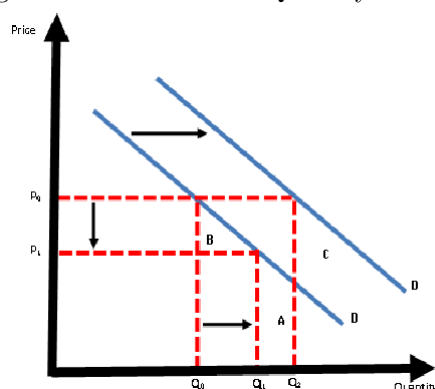
²This is one of the reasons I enjoy teaching. In all my years as a student of economics (which started at the dawn of time, back in 2003) I had never thought of saturation-level as one of the factors affecting demand of a good or a service. After all these years, I had someone taking ECO101 point out to me in class that saturation belongs in this list. And of course it does.

changes. However, look at the other six factors that we talked about. The price level remained the same. But at the same price level, because of a change in one of the factors (rise in income, for example), our quantity demanded went up. This is called a rise in demand (as opposed to rise in quantity demanded).

It is important that you notice that a change in demand is not the same thing as a change in quantity demanded.³ It is important that the difference between these two phenomenon are very clear to you. This, in my experience, is one of the most common mistakes made by students at the undergraduate level.

Price changes affect our buying decisions by changing our quantity demanded. Other factors affect our buying decisions by changing the quantity demanded at the same price – a change in demand.

Figure 3.1: Demand and Quantity Demanded



Refer to figure 3.1. Suppose the initial demand curve is D_0 . We are at point B, where price of the good is P_0 and quantity demanded at that price is Q_0 . If price goes down to P_1 , a higher quantity will be demanded, and we move from point B to point A, where $Q_D = Q_1$. Notice that we are still operating along the same demand curve – **a change in price of the good has led to a change in quantity demanded, which is represented by movement along the same demand curve**. This is called a change in quantity demanded.

However, suppose the price level does not change but there is a rise in income of the consumer(s). Due to a rise in income, at the same price of P_0 , more is demanded and we move to a new demand curve D_1 , and from point B to point C. Notice here that a change in a factor other than price has led to change in quantity demanded at the same price level (and at every price level), which is represented by a shift of the entire demand curve. This is called a change in demand.

Try to work out the analogous case of rise in price and decrease in demand.

3.2 Introduction to Supply

If we understand demand, understanding supply will be pretty straightforward. Intuitively, supply may be a trickier concept to understand for some. After all, we are all consumers,

³As you will see often, similar sounding words and expressions can have very different meanings in economics. ‘demand’ and ‘quantity demanded’ are different just like ‘price of a good’ and ‘price level’ are two different concepts (something you will learn about in a course more advanced than ECO101). It is important to be aware of these distinctions or you risk losing easy marks in your exams.

This is not unique to economics alone. If you have studied physics in school or at a more advanced level, you are already aware of the important distinction between ‘weight’ and ‘mass’ of an object even though they may sound similar.

Consumers want to buy a good or a service for the cheapest price possible. Producers want to sell a good or a service for the highest price possible.

purchasing various products every single day. As a result, understanding consumer behavior and motivations were easy for us. But not all of us are producers and getting inside the headspace of a producer may not come easily. But there is only one thing we need to keep in mind and everything else will fall in place. Price and quantity have a positive relationship for suppliers, as opposed to an inverse relationship that consumers have. That, effectively, is the crux of the conflicting motivations between buyers and sellers and if we can keep that in mind, we will have no difficulty in understanding producer theory. Let us get started.

Much like demand, there are three requirements to be fulfilled before we can say that a good or a service is being, will be, or can be, supplied to the market. The seller/producer/manufacturer must:

- Have the necessary resources and technology to make the good
- Be able to make a profit in the market
- Plan to produce and supply

The first requirement is obvious. If the producer does not have access to the resource and technology to produce the good, he is incapable of supplying the good to the market. Once he can do that, however, he needs to be able to make a profit at the market. If the production cost per unit is taka 10 but the good can only be sold for taka 7 in the market, there will be no supply in the market. And finally, with the requisite technology, resources, and profit-margin, the producer must also plan to produce the good and supply it to the market. Once we have all three, we have supply.

Since most of my examples have been about goods so far, let me give you an example about the supply of a service to the market. Let us talk about barbers who provide haircuts.

First of all, to supply haircuts to the market, the ‘supplier’ needs to be trained – he needs to have the ability to provide haircuts. He will also need a place where he can offer his haircuts; he will need combs and special hair-cutting scissors; and other equipment. These are the ‘resources and technology’ that he needs to provide haircuts. Once he has them, he needs to look at how much money he can charge for his services. There is a cost involved – his training cost, the cost of acquiring all the equipment, the rent of the place where he is cutting the hair – and if he cannot recover all these costs by charging an appropriate amount of money, he will not be providing haircuts to the market. And of course, even if the first two conditions are fulfilled, he may not be supplying haircuts to the market (think of retired barbers who probably satisfy the first two criteria but not the third one). Once he has concrete plans about when and where he will be providing haircuts, he is a barber and is supplying a service to the market.

Remember, a supplier wants to sell goods or services to the market at the highest possible price level.

The Law of Supply: A rise in the price of a good or a service will, ceteris paribus, lead to a rise in the quantity demanded of that good or service, vice versa.

Let us think a little more about what this says. There was an existing price level in the market based on which, a certain quantity of a good (or a service) was being supplied to the market. When that price level changed, and their profit-margins changed, suppliers responded to the change by adjusting the level of their supply. However, is price level the only thing that determines how much of a good is supplied to the market, or are there other factors that affect that decision?

3.2.1 Factors that Affect Supply

Beyond price level, what other factors affect supply decisions? There can be many factors but we have listed the most common and important factors underneath:

1. Cost/price of the factors of production

A production process can be said to be an input-output model. The producer introduces the inputs in the process – these includes labor/workers, raw material and resources, machinery and capital, et cetera – and in return, gets an output from the process – the final product. The inputs collectively are called the **factors of production**. Factors of production have associated costs – wage for workers, rent for the factory space, interest payment if machineries have been bought on loan. When one of these costs go up, the cost of production of the good or service goes up. This adversely affects the supply. The opposite is also true: a decline in the cost of production will increase supply.

Suppose a pen-maker spends taka 10 to make a pen and sell it at the market for taka 13. His profit per pen is taka 3. If now the wage rate of workers unexpectedly goes up and the pen-maker has to spend taka 12 to make a pen, his profit per pen falls to taka 1. With a lower profit margin, he will be less inclined to supply the old quantity to the market. Therefore, even though the price of pens has not changed in the market, the supply has fallen.

2. Price of related goods in production

This is an analogous case of the substitute and complement goods we talked about in our previous lecture about demands. During the production process, there are certain goods that are **substitutes in production** and others that are **complements in production**. Goods that are substitutes in production use the same raw material. Therefore, a rise in the price of one good will lead to a fall in the supply of the substitute (in production) good even though its price has not changed. Goods that are complements in production arise out of the same production process. Therefore, a rise in the price of one good will lead to the rise in supply of the other good.

A good example of two goods that are substituted in production is soft drinks and energy drinks. They both use almost the same ingredients and have similar packaging. So, if there has been a rise in the price of soft drinks, there will be an increase in quantity supplied of soft drinks. However, this diverts raw materials away from the production of energy drinks. As a result, the supply of energy drinks falls. Finding examples of complements in production is a little trickier but I think I have a good one. A rise in the price of beef will lead to an increase in the quantity supplied of beef. However, as more cows are sent to the abattoir to increase the quantity supplied of beef, more leather is produced in the process. Therefore, due to a rise in the price of beef, the supply of leather goes up.

3. Number of suppliers

This is pretty straight-forward. If we have more suppliers, there will be more supply in the market. However, we need to be careful when thinking about this. In the demand case, we saw how a higher population leads to higher demand, vice versa. There is no upper limit

to how high the population size can be. There may be ecological and demographic factors that prevent population from going higher than a certain limit, but within the purview of economics and economic theory, population size can increase to any number. The same cannot be said for the number of suppliers in market.

It is true that number of suppliers can be equated with the size of supply in the market. But remember, goods will only be supplied to the market if the producer expects to make a profit from selling the good in the market. Therefore, new suppliers will only enter the market as long as they think that there is unmet demand in the market that they can fill. Once all demand has been met, there is no motivation for additional supplier to enter the market and increase the supply. Therefore, there is an upper bound on the number of suppliers in any given market.

4. Technology

This of the effect better technology had on a PPF. It allowed us to produce more with the same amount of resources, which pushed out the boundary of the PPF. Any producer who experiences improved technology is able to produce the same goods but at a lower cost now. Therefore, there will be a rise in the supply.

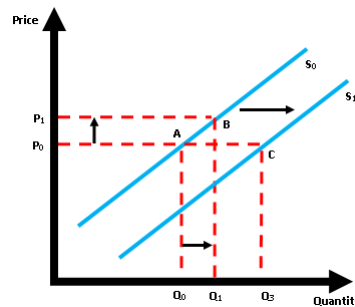
3.2.2 Supply and Quantity Supplied

The Law of Supply told us that whenever the existing price-level in the market changes, suppliers react to the change by adjusting how much they supply to the market – the quantity supplied changes. However, look at the other factors that we just discussed. The price level remained the same. But at the same price level, because of a change in one of the factors (rise in cost of factors of production, for example), the quantity supplied went down. This is called a fall in supply (as opposed to rise in quantity supplied).

Recall what I said at the end of our discussion about demand and quantity demanded. It is important that you notice that a change in supply is not the same thing as a change in quantity supplied. It is important that the difference between these two phenomenon are clear to you.

Price changes affect quantity supplied to the market. Other factors affect the quantity that is supplied at the same price – a change in supply.

Figure 3.2: Supply and Quantity Supplied



Suppose the initial supply curve is S_0 . We are at point A, where price of the good is P_0 and quantity demanded at that price is Q_0 . If price goes up to P_1 , a higher quantity will be supplied to the market, and we move from point A to point B, where $Q_S = Q_1$. Notice that we are still operating along the same supply curve – **a change in price of the good has led to a change in quantity supplied, which is represented by movement along the same supply curve.** This is called a change in quantity supplied.

However, suppose the price level does not change but there is a fall in labor cost (wage). Due to a fall in cost of production, at the same price of P_0 , more is supplied and we move to a new supply curve S_1 , and from point A to point C. Notice here that **a change in a factor other than price has led to change in quantity supplied at the same price level (and at every price level), which is represented by a shift of the entire supply curve.** This is called a change in supply.

Try to work out the analogous case of a fall in price and fall in supply.

3.3 Market Equilibrium

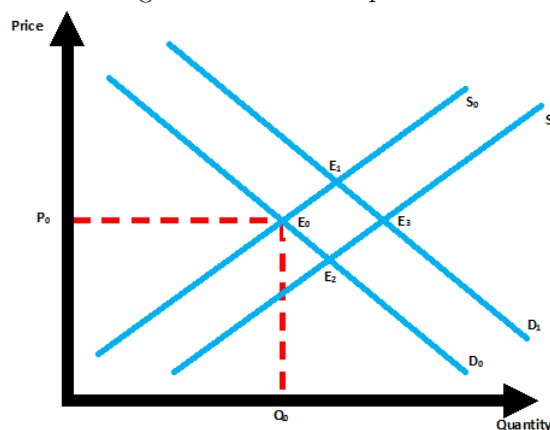
Let us think a little bit more about what we have learned during this and the last lecture and try to bring them together. We have learned what demand is and what factors affect the size of the demand of a good or a service from a market. Similarly, we have learned what supply is and what factors affect the size of the supply of a good or a service to a market. We have also seen a lot of fluidity – a multitude of factors that can change supply and demand. So is that how it always is? Is the market always in a flux of ever-changing sizes of demand and supply?

Not quite. Markets are surprisingly elegant constructs that are capable of stable outcomes. And markets achieve these stable outcomes without the need for any coordination between millions of consumers and thousands of producers. It is a truly remarkable phenomenon when you think about it.⁴ This phenomenon, in economics, is known as **market equilibrium**.

So far, we have talked about two **economic agents**: the consumers and the producers. Therefore, a market equilibrium will be the point at which, given the price level, neither the consumers nor the producers have any motivation to either increase or decrease the quantity that they are consuming from or supplying to the market. In more technical terms, market equilibrium is the point at which, at a given price level, the quantity supplied to the market is exactly equal to the quantity demanded from the market. Graphically, this is the intersection-point between the positively-sloped supply-curve and the negatively-sloped demand-curve.

Equilibrium is a point at which there is no further motivation for anyone involved to change anything.

Figure 3.3: Market Equilibrium



With an initial demand curve D_0 and supply curve S_0 , the intersection point, E_0 is the

⁴If you are interested, you may want to find out more about the *laissez-faire* approach and the concept of the *invisible hand* of the market introduced to us by the famous Scottish moral philosopher Adam Smith in 1776.

market equilibrium. That means that at price P_0 , quantity demanded equals quantity supplied, which is Q_0 .

If either demand or supply were to increase (or decrease), we would have a new equilibrium point. For example, if demand increased due to economic growth and a rise in income level of the consumers, demand curve would shift to the right, from D_0 to D_1 . This takes us to the equilibrium point, E_1 . At the new equilibrium, both price and quantity higher.

What would happen to the equilibrium if supply decreased? What would happen if demand decreased and supply increased?

The next question is, how do we achieve equilibrium, E ? What are the things that thousands of producers and millions of consumers have to do in coordination, to ensure that markets are in equilibrium? Nothing. We do nothing.

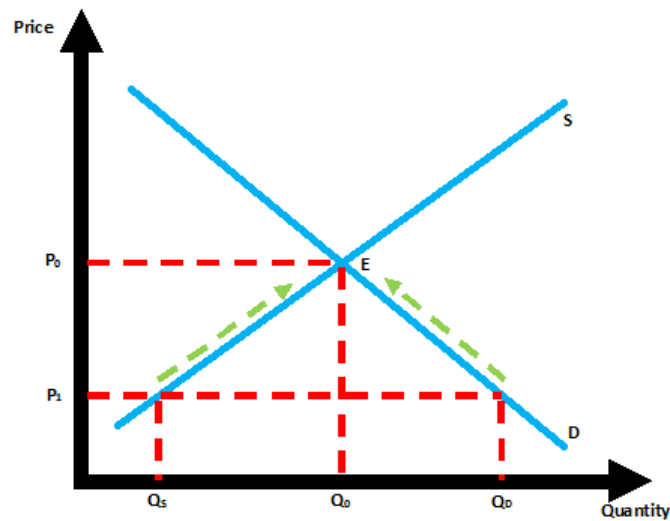
We leave it up to the markets to take care of themselves.

I did not say that markets are surprisingly elegant in how they achieve stable outcomes frivolously.

3.3.1 Shortages, Surpluses, and Price Adjustment

Mismatches in the quantity supplied and the quantity demanded in the market often occurs. It is only to be expected, since millions of individual agents participate in each market, pursuing their own personal goals. You put a million people in one room and ask them to make one simple decision and you will have pandemonium and chaos. Put even as few as 10 people in a conference room and ask them to make the most basic business decision and the meeting may drag on for hours, conflicts may break out, and the decision eventually will have to be made through a vote, not a unanimous agreement. Then how does a market, an intangible theoretical construct after all, manage to eliminate mismatches and give us equilibrium prices and quantities?

Figure 3.4: Price Adjustment



In the diagram above, D and S are the demand and supply curves and E is the market equilibrium. But suppose, instead of the equilibrium price P_0 , the price at the market is P_1 ,

which is lower than P_0 . Due to this lower price, quantity demanded is higher and quantity supplied is lower than the equilibrium quantity, such that $Q_D > Q_0 > Q_S$.

The mismatch between quantity demanded and quantity supplied ($Q_D - Q_S$) is the excess demand amount in the market (this can also be called a shortage in supply).

How can we go from here to the equilibrium point?

We simply need to think about how the two groups in the market, consumers and producers, will react to this scenario. Producers will see that there is excess demand in the market and they have a chance to increase supply and sell at a higher price. Some of the consumers will see that their demands are not being met at the market and stop demanding. Both these reactions will reduce the mismatch between quantity demanded and quantity supplied, and push the price up.

It may not happen immediately, but eventually, as enough producers increase their supply and consumers leave the market, quantity demanded and quantity supplied matches.

3.3.2 Mathematical Equations for Demand and Supply Curves

A negatively-sloped demand curve can be represented with the following equation: $P = a - bQ_D$, where a and b are real-numbers, P is the price level and Q_D is quantity demanded. Similarly, a positively-sloped supply curve can be represented with the following equation: $P = c + dQ_S$, where c and d are real-numbers, P is the price level and Q_S is quantity supplied. To find the equilibrium price and quantity, we set the two equations as equal, so that $a - bQ = c + dQ$. Solving this gives us the equilibrium quantity, $Q^* = \frac{(a-c)}{(b+d)}$. If we take this equilibrium quantity value and plug it into either one of the demand or supply equation, we will get the equilibrium price level, $P^* = \frac{(ad+bc)}{(b+d)}$.

Numerical Example

Suppose we have the following:

Demand: $P = 400 - 2Q_D$ and Supply: $P = 100 + 1Q_S$

We can also write these functions as $Q_D = 200 - \frac{1}{2}P$ and $Q_S = P - 100$

At the equilibrium, quantity demanded and quantity supplied are the same at a given price level. Therefore:

$$400 - 2Q_D = 100 + 1Q_S \Rightarrow Q^* = 100$$

To find the equilibrium price level, we plug in the value of Q^* into one of the equations. Suppose we choose the supply curve. Therefore, we get:

$$P^* = 100 + 1(100) = 200$$

Therefore, for the given supply and demand, the market reaches its equilibrium when 100 units of the good are traded in the market at a unit price of 200 each.

3.3.3 Economic Analyses Using these Equations

Let us jump right into a numerical example.

Numerical Example

Suppose we are dealing with the same market as our previous example. Therefore, Demand: $Q_D = 200 - \frac{1}{2}P$ and Supply: $Q_S = P - 100$. We also know that $Q^* = 100$ and $P^* = 200$.

To help us understand better, let's assume that these are the demand and supply functions of a set-menu at your University cafeteria. This is the only set-menu available in the cafeteria, it is priced at taka 200 and sells 100 units every day during lunch. Now suppose that other items from the menu become cheaper. Given that they are substitutes for the set-menu, there is an immediate fall in the demand due to a fall in the price of a substitute good.

Suppose that demand function decreases by 50, meaning that it shifts to the left by 50 units. Therefore, the new demand function, $Q_D = 150 - \frac{1}{2}P$.

Now here is the tricky part: do you think that the equilibrium quantity will fall by 50 units, so that $\hat{Q}=50$? Not quite. This fall in demand certainly means that at the same price level, 50 fewer units are demanded. But now, we have a mismatch – at price level 100, consumers have a demand of 50 units in the market but suppliers are willing to supply 100 units to the market – there is excess supply to the market (or in this case, the cafeteria).

We know what will happen in response to this. Since all units are not being sold, suppliers will reduce the quantity they supply to the market leading to a fall in price level which will increase the quantity demanded to a level above 50. Let us calculate how much.

New demand curve: $Q_D = 150 - \frac{1}{2}P$ and the supply curve has remained unchanged: $Q_S = P - 100$

Therefore, we can write $150 - \frac{1}{2}P = P - 100 \Rightarrow \hat{P}=167$.

Plugging in this value of equilibrium price level into either one of the equations, we get, $\hat{Q}=67$

Therefore, what we see is that due to a fall in the price of substitutes, the demand for the set-menu falls, taking us to a new equilibrium of $\hat{Q}=67$ and $\hat{P}=167$.

I will recommend that you try to do this graphically. Start off by plotting the initial demand and supply curves and find the equilibrium point. Then shift the demand curve by 50 units and find the new equilibrium point. See if you understand, step-by-step, how what decisions are taken by the consumers and suppliers and how that changes the equilibrium values.

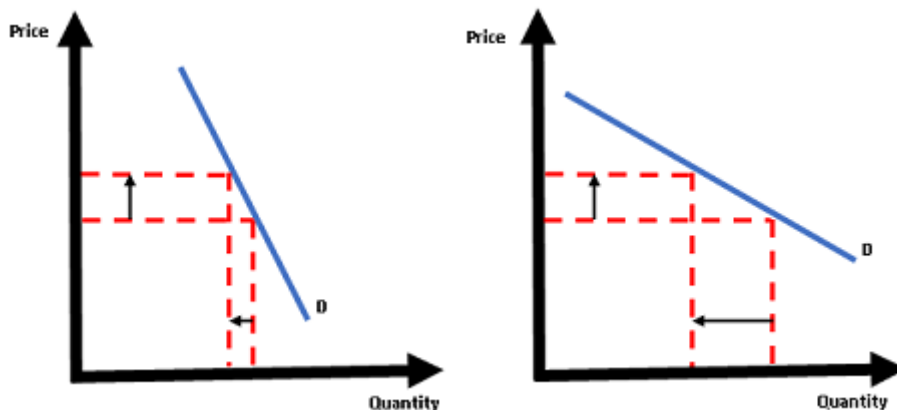
And there you have it. A simple economic analysis, conducted using the basic economic tools that we have learned so far. We will return to market equilibrium again later in our course. There are still some very interesting aspects that arise out of a market equilibrium that we are yet to discuss. However, there are few other topics that we need to understand before we can do that. In our next lecture, we will start our talks about what was my favorite topic of economics back in school – elasticity.

Chapter 4

Elasticity

So far, we have only focused on the direction of the relationship between two variables. For example, the law of demand gives us a negative/inverse relationship between variables and the law of supply gives us a positive relationship. We should now turn our focus on the ‘magnitude’ of these relationships. Basically: when talking about two variables and one variable changes, we want to know in what direction will the other variable move, and *by how much*. This phenomenon is called the responsiveness – or the **elasticity** – of one variable in response to a change in another.

Figure 4.1: Elasticity



Both goods experience the same price rise, from P_0 to P_1 . However, the fall in quantity demanded experienced in the two markets are very different. That is because the two goods have two different elasticities of demand. What factors govern these relationships?

4.1 Price Elasticity of Demand

There are different types of elasticity and we will talk about a few of them over the course of this semester. Let us start with the most common one, the **price elasticity of demand (PED)**.

For the first time in this course, I am going to introduce a formula. You will of course need to learn this formula to be able to answer questions in your quizzes and exams. But understanding is always better than rote-memorization. If you understand what exactly we are

The price elasticity of demand measures the responsiveness, or the elasticity, of the quantity demanded of a good or a service to changes in its price.

trying to measure through elasticity, you will understand the formula in a way that precludes the need for memorization. Take another look at the formula. PED is the change in quantity demanded due to a change in price. The formula is:

$$PED = \frac{\% \Delta \text{ in Quantity Demanded of } X}{\% \Delta \text{ in Price of } X} = \frac{\Delta Q / Q_{avg}}{\Delta P / P_{avg}}$$

Notice that the formula is different from the formula that you may be familiar with to calculate the percentage change in a variable ($\% \Delta = \frac{(new-old)}{old}$). We use the modified formula to make sure that we get the same elasticity if price rises from P_1 to P_2 or if price falls from P_2 to P_1 .¹

Please keep in mind that PED along a straight-line is not constant. Verify this on your own.

It should also be obvious to you that the PED is almost always negative, since price and quantity demanded moves in opposite directions. Since it is always negative, the sign in front of the number is inconsequential for our analyses. To measure the responsiveness or the elasticity of the variables, we are solely interested in the size/magnitude of the calculated value. In a nutshell, we care more about degree and not so much about the direction. If your calculated PED is -5, it is the same to say that the PED is 5. Therefore, let us ignore the sign for now and focus on the number.²

Keep in mind that PED is always negative. Price elasticity of demand can take the following values:

1. $PED = 0$: *Perfectly Inelastic Demand*

Regardless of changes in the price level, quantity demanded will not change at all. Think about the formula. Regardless of the value in the denominator, the value in the numerator remains 0. This applies for necessary goods. Two good examples of perfectly inelastic demand are insulin and student ID cards. Graphically, goods with perfectly inelastic demands have vertical demand-curves.

2. $0 > PED > -1$: *Inelastic Demand*

An $x\%$ change in the price level leads to a $y\%$ change in the quantity demanded, given that $x > y$. This applies for important goods, such as food and housing.

3. $PED = -1$: *Unit Elastic Demand*

An $x\%$ change in the price level leads to a $y\%$ change in the quantity demanded, as long as $x = y$.

4. $PED < -1$: *Elastic Demand*

An $x\%$ change in the price level leads to a $y\%$ change in the quantity demanded, given that $x < y$. This applies for goods such as cars and furniture.

5. $PED = -\infty$: *Elastic Demand*

A small rise in the price level will take quantity demanded to zero. This applies for goods with very close substitutes. For example, two vendors selling vegetables from a cart on either sides of a street will face perfectly elastic demand: if one of them tries to charge even a slightly higher price, he loses all his customers to the competitor. Graphically, goods with perfectly elastic demands have horizontal demand curves.

¹However, I will still accept the original formula if you choose to use it

²Also known as the absolute value. For example, $|a| = |-a| = a$

Numerical Example

Suppose you are at your local pharmacy to buy these two things – one Sultolin inhaler and one strip of Paracetamol. Sultolin is something you take for your respiratory/breathing problems. You need to take at least 2 puffs every day or it becomes very difficult for you to breathe normally. You take Paracetamol because of some minor headaches or high temperature that you get once every few weeks.

When you show up at the pharmacy, we find that the prices of both the goods have gone up. Suppose, you go through 10 Sultolin inhalers in a year. This is not a medicine you can function without. The shop-keeper tells you that there is a cheaper local alternative available, called Asmasol, but you do not want to take the risk. Similarly, suppose you buy about 10 strips of Paracetamol tablets every year. You are well aware that there are other alternatives available for Paracetamol and you do not need to buy Paracetamol if it is expensive.

Therefore, we have the following values:

Old price of Sultolin, $P_S^1 = 250$

New price of Sultolin, $P_S^2 = 300$

Old quantity purchased of Sultolin, $Q_S^1 = 10$

New quantity purchased of Sultolin, $Q_S^2 = 9$

Old price of Paracetamol, $P_P^1 = 10$

New price of Paracetamol, $P_P^2 = 11$

Old quantity purchased of Paracetamol $Q_P^1 = 10$

New quantity purchased of Paracetamol, $Q_P^2 = 6$

Price elasticity of demand (PED) of Sultolin: $\frac{\% \Delta \text{in } Q_S}{\% \Delta \text{in } P} = \frac{\frac{1}{9.5}}{\frac{50}{275}} = 0.58$

Price elasticity of demand (PED) of Paracetamol: $\frac{\% \Delta \text{in } Q_P}{\% \Delta \text{in } P} = \frac{\frac{4}{8}}{\frac{1}{10.5}} = 5.25$

We get the results we would expect to. Since Sultolin is an important medicine (maybe even necessary), and does not have many good substitutes, it has a low PED, at 0.58. That means that a 1% change in the price level will lead to a 0.58% change in the quantity demanded of Sultolin. On the other hand, Paracetamol is not a necessary medicine for you and it has a few close substitutes available. Therefore, it has a high PED of 5.25, meaning that a 1% change in the price level will lead to a 5.25% change in the quantity demanded.

There are three factors that influence the elasticity of demand:

1. **Availability of close substitutes:** If a close substitute exists, then even a small rise in the price will see a large fall in the quantity demanded, vice versa. As a result, it is obvious why Pepsi has a very high PED and insulin³ has a very low PED.
2. **Portion of income spent on the good:** Goods that cost a small portion of our total budget have small PEDs since we can afford the price hike. However, for goods that constitute a relatively larger portion of our total budget, a small price rise may make them unaffordable and therefore they will have a high PED.

³You may want to argue that insulin has a low PED because it is a necessary good, not because it has no close substitutes. You would be correct in your assessment. However, it can also be said that insulin is necessary because it has no close substitutes. If there was an alternative to insulin, it would no longer be necessary. Patients could just buy the other product. At the end of the day, both factors are valid.

If salt prices went up by 25%, people would continue to buy around the same amount of salt because it constitutes a small part of our total budget and we can afford to pay a little extra. However, if meat prices were to go up by 25%, we would have to reduce our demand for meat by a large amount. And if car prices go up even by 5%, we may no longer be able to afford to buy the car we have been planning to and our demand falls to zero.

3. **Time elapsed since the price change:** It is not always possible to change our consumption patterns immediately. Even if there has been a price hike, we may have to continue with our existing consumption-mix in the short-run, and only be able to change it in the long-run. Therefore, immediately after a price change, goods can have low PED, but over time, PED will be higher.

If rice becomes expensive suddenly, we will still need to consume rice. We cannot suddenly reduce our consumption of rice, which constitute as the largest portion of carbohydrate (and calorie) intake in Bangladesh. If fuel prices were to suddenly go up, we will still need to continue buying fuel to run our cars, our factories, et cetera. However, over time, if price level remains high, it is possible that we may be able to come up with other alternatives – maybe we start having more bread and pasta instead of rice and maybe we develop alternate or renewable energy technologies that reduce our dependency on fuel.

4.1.1 Price Elasticity of Demand and Total Revenues

PED is an important tool of analysis for producers as it tells them how consumers in the market will react to a change in the price level.

Knowing the PED goes a long way in helping producers, manufacturers, and/or sellers decide whether to increase the price level or not. This should intuitively make sense. If a good has inelastic PED, a rise in the price level will lead to a relatively smaller fall in the quantity demanded. Therefore, if producers increase the price by 10%, quantity demanded will fall, but by less than 10%. Therefore, by increasing the price, producers can increase their total revenue (total revenue, $TR = \text{Price} \times \text{Quantity Sold}$). The reverse is also true: with elastic PED, a rise in price level will lead to a fall in total revenue and in these cases, producers should not raise the price level. If they do, they will lose money. It is better to lower prices in these instances which will increase total revenue.

Economics as the Dismal Science

Scholars today widely consider Adam Smith to have started the school of modern economics with his 1776 book, *The Wealth of a Nation*. Smith was followed by other great economists in the 19th century who formalized the discipline and developed the foundations of ideas that we still find useful today. However, by 1849, Thomas Carlyle had termed economics as the dismal science. Many have called economics far worse since. There are many reasons why but the preceding paragraph is one of the best instances.

The lesson from the paragraph is simple: increase prices if elasticity is low and decrease prices if elasticity is high.

But what are the implications of this lesson? Raise the price level of necessary goods such as medicine and foods and you will become rich because people *have to* buy them. Lower prices of soft drinks and other fast-food and more people will consume these unhealthy, demerit products and once again, you will make lots of money.

Now this is not a part of your course syllabus. But I want you to think about this for a little while. Maybe have an extended discussion with your friends and acquaintances. Think about all that you have learned so far and the things you will learn for the rest of the course. What is your opinion of economics? Do you think it will help you make decisions in your life?

A very good place to start can be by trying to answer this simple question: is economics a descriptive or a prescriptive discipline to you?

4.2 Cross Elasticity of Demand

We are going to continue our discussion on elasticity. Remember, you can create elasticities of anything two related things. For example, you may want to create a 'hours studied elasticity of CGPA' to find out by what percentage your CGPA will increase by if you study for 1% longer. However, for this course, we are going to focus on four important elasticities. We have already discussed one of them, the price elasticity of demand.

The second elasticity we will talk about is the **cross elasticity of demand (XED)**.

The formula to calculate XED:

$$XED = \frac{\% \Delta \text{ in Quantity Demanded of good A}}{\% \Delta \text{ in Price of good B}} = \frac{(\frac{\Delta Q}{Q_{avg}}) \text{ of good A}}{(\frac{\Delta P}{P_{avg}}) \text{ of good B}}$$

Cross elasticity of demand measures the responsiveness, or the elasticity, of the demand of one good, or a service, to changes in the price of another good, or service.

The important thing to notice here is that XED measures the nature of relationship between two goods. We know that two goods can have three types of relationships: they can be substitutes, complements, or be unrelated. These relationships will be important in our analyses of XED. XEDs can take on the following values:

1. $XED > 1$: Elastic demand; substitute goods

Going back to the formula, we can see that if both the numerator and denominator are positive (or negative), then good A and good B are substitutes. A rise in the price of one good leads to a rise in the demand of the other good, vice versa. The high value of XED (more than one) indicates that the goods have an elastic relationship – they are **close substitutes**. An x% change in the price of good B will lead to a more than x% change in the demand of good A.

2. $XED \in (0, 1)$: Inelastic demand; substitute goods

Same as above, only, the relationship is inelastic now. Therefore, an x% change in the price of good B will lead to a less than x% change in the demand of good A. These are **weak substitutes**.

3. $XED = 0$: No relationship

A change in the price of one good has no effect on the price of the other good. They are not related.

4. $XED \in (-1, 0)$: Inelastic demand, complement goods

If XED is negative, good A and good B are complements (refer to the formula if you are not sure why). A rise in the price of one good leads to a fall in the demand of the other good, vice versa. However, these are **weak complements**. An x% change in the price of good B will lead to a less than x% change in the demand of good A.

5. $XED < -1$: Elastic demand; complement goods

Same as above, only, the relationship is elastic now – these are **strong complements**. An x% change in the price of good B will lead to a more than x% change in the demand of good A.

Coke and Pepsi are strong substitutes and will have an elastic XED. A small rise in the price of one of them, Coke for example, will lead to a relatively large increase in the demand of Pepsi, vice versa. On the contrary, tea and coffee are substitutes, but very weakly. When you want coffee, you want coffee, and very rarely will you be happy about having to substitute one for another.

A change in price of shirts will leave the demand of phones unchanged – these goods are unrelated and have a 0 XED.

Pencils and sharpeners are strong complements since you have to use them together. A rise in price of one will see similar falls in demands of both goods. However, pencils and erasers are weaker complements. They are still used together, but it is possible to use a pencil without the need of an eraser.

Unlike the PED case, we have already discussed the factors that affect the XED between two goods. They are:

- The nature of relationship between the goods (substitutes of complements)
- The relative strength or weakness of the relationship

4.3 Income Elasticity of Demand

Income elasticity of demand measures the responsiveness, or the elasticity, of the demand of a good, or a service, to changes in the income level of the consumer.

The third type of elasticity we will talk about is the **Income Elasticity of Demand (YED)**. The formula to calculate YED:

$$YED = \frac{\% \Delta \text{ in Quantity Demanded}}{\% \Delta \text{ in Income}} = \frac{\left(\frac{\Delta Q}{Q_{avg}} \right)}{\left(\frac{\Delta Y}{Y_{avg}} \right)}$$

YED measures the nature of relationship between income and the demand of a good. We know that this relationship can be of two types: a positive relationship (normal goods) or a negative relationship (inferior goods). YEDs can take on the following values:

1. $YED \in (0, 1)$: Inelastic, normal

A positive relationship between quantity demanded and income tells us that the good in question is normal. In this case, we are talking about inelastically normal goods – an x% change in income leads to a less than x% change in the quantity demanded.

2. $YED > 1$: Elastic, normal

A positive relationship between quantity demanded and income tells us that the good in question is normal. In this case, we are talking about elastically normal goods – an x% change in income leads to a more than x% change in the quantity demanded.

3. $YED < 0$: Inferior

A negative relationship between quantity demanded and income tells us that the good in question is inferior. Of course, we can extend our discussion here and talk about elastically and inelastically inferior goods. However, it is rare that such analyses are conducted.

4.4 Price Elasticity of Supply

The fourth and final type of elasticity we will talk about is the **Price Elasticity of Supply (PES)**. The formula to calculate PES:

$$PES = \frac{\% \Delta \text{ in Quantity Supplied}}{\% \Delta \text{ in Price}} = \frac{\left(\frac{\Delta Q}{Q_{avg}} \right)}{\left(\frac{\Delta P}{P_{avg}} \right)}$$

PES measures the nature of relationship between the price of a good or a service and its supply. We can tell from the Law of Supply that this relationship is always positive. PES, in addition, allows us to talk about the degree/magnitude of this relationship. Two factors affect the size of PES:

Price elasticity of supply measures the responsiveness, or the elasticity, of the supply of a good, or a service, to changes in the price of the same good.

1. **Time-frame:** Goods that take a long time to produce (for example, agricultural products) have low PES since they cannot react quickly to a rise or fall in price. Other goods that can quickly and easily be made (small manufactured goods such as pens and markers) have high PES since they can be highly responsive to changing prices.
2. **Resource substitutability:** If production of a good requires the use of a resource not easily available, that good will have a low responsiveness to price changes. ‘Resources’ can be a specific raw material, workers with a special skill-set, a particular type of machinery, et cetera. In each case, producers will face difficulty in quickly raising supply if the needed resource is not easily available or cannot be scaled-up to the requisite level.

Chapter 5

Equity and Welfare

5.1 Benefit, Willingness to Pay, and Consumer Surplus

We are going to start our discussion on this new topic by first differentiating between *value* and *price*: **value is what we get and price is what we pay**. Keep in mind that price is set by the market and we have no control over it individually. Every consumer pays the same price. Value, however, is subjective from individual to individual and varies widely.

You may be a huge Marvel fan and have been willing to pay a high price to get your hands on a ticket to one of the first premiers of Avengers: Endgame. Suppose you are willing to pay taka 2,000 for an early ticket. Your perceived value of watching the movie was high. I, as a casual fan, was willing to pay slightly higher than what I usually pay for movie tickets, maybe around taka 600, but not more – my valuation of watching the movie was much lower.

However, that is only what we are willing to pay and has no bearing on the amount we will actually have to pay. When we went to the ticket-counter, the attendant charged us both the same price – taka 400. He did not know that we valued the movie differently, and even if he had known that, he would have still charged us the same price – the price in the market is applicable for all.

After watching the movie, I find myself satisfied with the transaction – I considered the taka 400 well spent and went home. But you, as a big fan, wanted to watch the movie again immediately. Having seen the movie once already, you were no longer willing to pay taka 2,000, but you would have gladly paid taka 1,000 for a second screening. When you went to the ticket-counter, how much were you charged? Less, because your willingness to pay (WTP) is now lower? More, because you have proved yourself to have a high demand by going back?

No, you would have paid the exact same amount again – taka 400!

And you could have continued to watch the movie as many time as you wanted until your WTP went under taka 400. In that case, your WTP is lower than the price of a ticket and you no longer see yourself benefitting enough from the transaction to spend the money.

Your **willingness to pay (WTP)** is directly related to your **marginal benefit (MB)**. You are willing to pay an amount that is equal to the benefit you will derive from consuming the next

unit – your marginal benefit.

After a day out in the sun, you may find yourself so thirsty, that you are willing to pay taka 100 for a small bottle of Mum. Why is that? Because in that thirsty state, your benefit from consumption of the first bottle of Mum water – your MB of a bottle of Mum water – is very high. You are willing to pay taka 1,000 because you value your benefit from consuming the first bottle of Mum water to be worth taka 1,000.

But after having consumed the first bottle, when you purchase the second bottle, you are willing to pay less – because the benefit you will derive from consuming the second bottle – the MB of the second bottle of water – is low. The first bottle may have saved you from suffering a heatstroke but the second bottle only helps you quench your thirst (if you are still thirsty) and the third bottle only works as a safety-measure that you put in your backpack in case you get thirsty again in a few hours – with each subsequent purchase of a bottle of water, their marginal benefit falls as does your willingness to pay.

Your marginal benefit curve is your marginal willingness to pay curve is your individual demand curve. At price taka 10, you are willing to buy 5 units of the good, that means that you value the 5th unit as giving you a marginal benefit worth taka 10. When the Law of Demand tells us that with a rise in price, the quantity demanded falls, it basically means that if we curtail our consumption to a lower level, we are willing to pay a higher at the margin.

The **market demand curve** is the horizontal sum of the individual demand curves and is formed by adding the quantities demanded by all the individuals at each price. Analogues to how your individual demand curve is your marginal benefit curve, we can also say that **the market demand curve is the marginal social benefit (MSB) curve**. We will come back to the MSB curve in our next lecture.

We do not always have to pay what we are willing to. If what we are willing to pay is less than the market price, then we will not buy the good since we see no value in the transaction.¹ Then, it goes to reason, that we will only buy in the market if our willingness to pay is exactly equal to (rarely) or more than (often) the market price. This additional benefit that we accrue from the market without having to pay for is called the **consumer surplus**.

Usually, we can calculate total consumer surplus (for an individual or the entire society) by calculating the area of the triangle between the price line, the demand curve, and the y-axis. I assume you all know how to find the area of a triangle.

If consumers enjoy consumer surplus from every transaction², what do producers enjoy? There is such a thing called the producer surplus. But to discuss that, we first need to talk a bit more about supply and marginal costs. We will do that in our next lecture.

Consumer surplus is the excess benefit received from a good over the amount paid for it, or in other words, the benefit that we do not pay for.

¹Will we ever have negative consumer surplus? Reconsider the example. When the price of print-outs exceeded the marginal willingness to pay, we decided not to make the transaction – we will never purchase a good if the cost (price) is higher than our willingness to pay. Therefore, consumer surplus for an individual will either be positive, or zero.

²Although, things may not always seem that way. In fact, people often complain that things are too expensive and one has to pay far higher than what he or she will willing to pay. Well, that is quite a fatuous thing to say. *By the very act of paying for it, you have just proved that you are willing to pay!* I think what people mean to say is that they would prefer if prices were lower. But that would be a true statement, no matter how low prices went.

The words we use to express ourselves are very important in a professional environment, and especially so when we talk about economics. You never pay an amount you are unwilling to pay for. After all, if your medicines are too expensive, you can always not take the medicines and suffer (or die). And if your house rent is too high, you can always move into the smaller house or roam the streets like a hobo. The fact that you do not do any of that proves that you are willing to pay the high price that you take so much delight in complaining about.

5.2 Cost, Minimum-price Supply, and Producer Surplus

Just like we can distinguish between *price* and *value* for consumers, we can distinguish between *price* and *cost* for producers. **Cost is what a producer gives up and price is what he receives.** The cost of producing one more unit of a product is its **marginal cost (MC)**. MC is the minimum price that a producer must receive to produce/sell another unit of the good. If the price is lower than the MC (his cost of production), then obviously he will not produce that good. An MC curve is a supply curve.

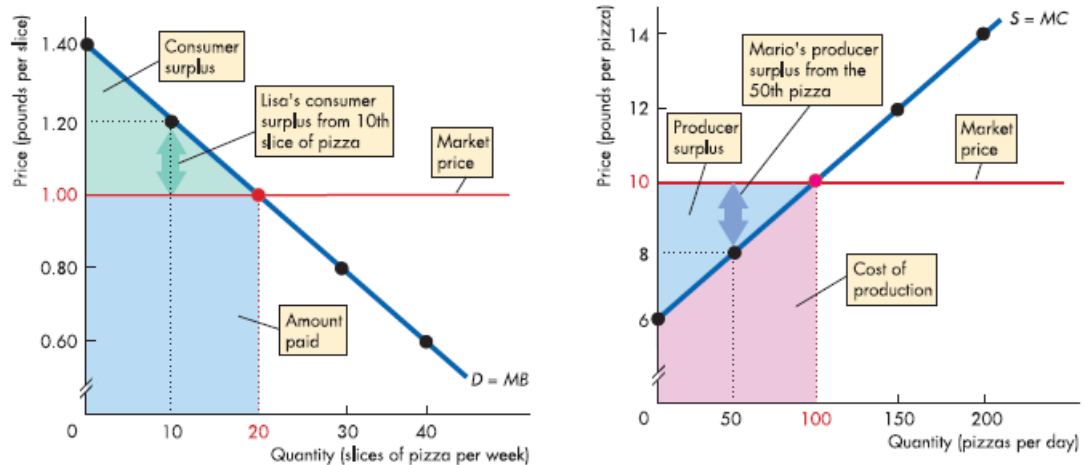
The market supply curve is the horizontal sum of the individual supply curves and is formed by adding the quantities supplied at each price level. **Market supply curve is also called the Marginal Social Cost (MSC) curve.**

Producers don't always get only the cost of their production, their marginal cost. They often receive more from the market. This is called a **producer surplus**.

Will there ever be negative producer surplus? It is unlikely that it will be. If a producer sees that his marginal cost of the next unit is higher than the amount he will receive by selling it in the market ($MC > P$), he will not produce that unit.

Producer surplus is the price of a good or a supply in excess of its marginal cost – the benefit received by the producer that he had not incurred.

Figure 5.1: Consumer and Producer Surplus



Lisa's demand curve for pizza tells us that at £1.40 a slice, she does not buy pizza. At £1.20 a slice, she buys 10 slices a week; at £1 a slice, she buys 20 slices a week. Lisa's demand curve is also her marginal benefit curve, so her demand curve also tells us that she is willing to pay £1.20 for the 10th slice and £1 for the 20th slice. She actually pays £1 a slice – the market price – and buys 20 slices a week. She spends £20 a week on pizza – the area of the blue rectangle. Lisa's consumer surplus from pizza is £4 – the area of the green triangle.

Mario's supply curve of pizza tells us that at a price of £6, Mario plans to sell no pizzas. At a price of £8, he plans to sell 50 pizzas a day; and at a price of £10, he plans to sell 100 pizzas a day. Mario's supply curve also tells us that the minimum he must be offered is £8 for the 50th pizza a day and £10 for the 100th pizza a day.

If the market price is £10 a pizza, he sells 100 pizzas a day and receives £1,000. The red area shows Mario's cost of producing pizza, which is £800 a day, and the blue area shows his producer surplus, which is £200 a day.

5.3 Market Efficiency

Now that we have discussed marginal benefit and marginal cost, we can ask an important question: do market equilibriums represent efficient outcomes? Or, in other words, are the market equilibriums the best we can do, or are there other better outcomes at a different price level and quantity?

The answer is no – a free-market equilibrium is the best outcome for both producers and consumers. Market equilibrium represents a point where marginal benefit of consumers and marginal cost of producers is equal. If we take a point on the left of the equilibrium point, $MB > MC$, meaning that goods are valued at higher than their cost. Therefore, consumers want to consume more and producers want to produce more. Any point on the right of the equilibrium point, $MC > MB$, meaning that the cost of producing goods is higher than their valuation. Consumers want to consume less and producers want to produce less. Only at the equilibrium point we have $MB = MC$.

Another way of approaching this problem is to look at the **total surplus**. On the left of the equilibrium point, we are not maximizing total surplus that we can by increasing Q . Similarly, on the right of the equilibrium point, we are not maximizing the total surplus (since we have negative consumer and producer surpluses) that we can do by decreasing Q . At the equilibrium point, both consumer and producer surpluses are maximized.

$$\text{total surplus} = \text{consumer surplus} + \text{producer surplus}$$

Refer to figure 5.2. Equilibrium point is given by E . Consumer surplus is the area of the yellow triangle and producer surplus is the area of the green triangle. At any point on the left of the equilibrium, E , $Q < Q_0$, we see that $MB > MC$ and both consumers and producers want to increase Q . At these points, the full benefits of consumer and producer surpluses have not been fully realized.

At any point on the right of the equilibrium, $Q > Q_0$, we see that $MC > MB$ and both consumers and producers want to decrease Q . At these points, we have negative consumer and producer surpluses and they can be removed by decreasing Q .

At point E , $MB = MC$, and total surplus is maximized.

At this equilibrium point, the marginal social benefit is equal to the marginal social cost. This condition is called **allocative efficiency**.

5.3.1 Market Failures and Deadweight Loss

Markets, however, do not always achieve an efficient outcome. When the market delivers an inefficient outcome, we call it **market failure**. Whenever there is overproduction or underproduction in the market, we measure the level of inefficiency by measuring the **deadweight loss**. Higher the inefficiency in the market outcome, higher the deadweight loss.

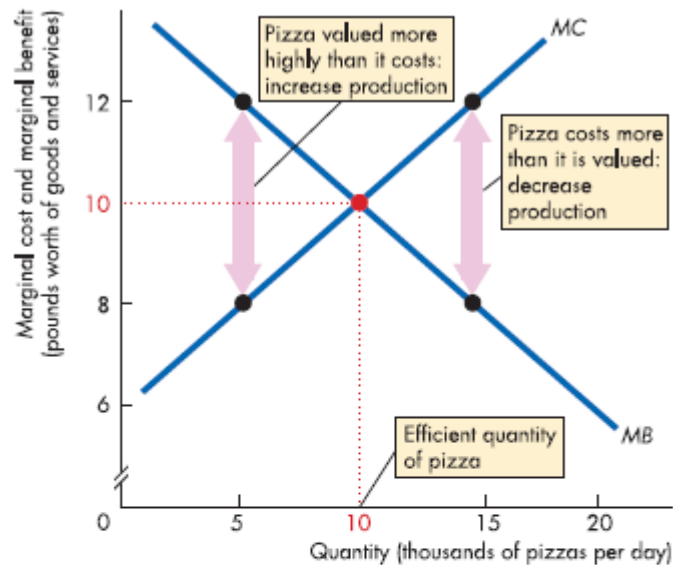
Refer to figure 5.3. If pizza production is cut to only 5,000 a day, a deadweight loss (the grey triangle) arises in part (a). Consumer surplus and producer surplus (the green and blue areas) are reduced. At 5,000 pizzas, the benefit of one more pizza exceeds its cost. The same is true for all levels of production up to 10,000 pizzas a day.

If production increases to 15,000 pizzas a day, a deadweight loss arises in part (b). At 15,000 pizzas a day, the cost of the 15,000th pizza exceeds its benefit. The cost of each pizza above 10,000 exceeds its benefit. Consumer surplus plus producer surplus equals the sum of the green and blue areas minus the deadweight loss triangle.

There are various reasons why markets may fail but they all boil down to one key reason. In a nutshell, markets fail when individual agents fail to include all the benefits and costs of their actions in their decision-making process. For example, pollution leads to considerably high social costs. However, individual producers may not consider it as a part of their marginal cost.

Market failures occur because too little of a good or a service (underproduction) or too much (overproduction) is being produced. Deadweight loss is the loss/decrease in total surplus that arises from an inefficient level of production.

Figure 5.2: Market Efficiency



The marginal benefit curve (MB) shows what people *are willing to forgo* to get one more pizza. The marginal cost curve (MC) shows what people *must forgo* to get one more pizza. If fewer than 10,000 pizzas a day are produced, marginal benefit exceeds marginal cost. Greater value can be obtained by producing more pizzas. If more than 10,000 pizzas a day are produced, marginal cost exceeds marginal benefit. Greater value can be obtained by producing fewer pizzas. If 10,000 pizzas a day are produced, marginal benefit equals marginal cost and the efficient quantity of pizza is available.

This means that they underestimate their marginal cost of production, leading to overproduction (high equilibrium Q) and lower price of the good than the socially optimum level. (Graphically, lower MC means the MC/supply curve shifts downward.)

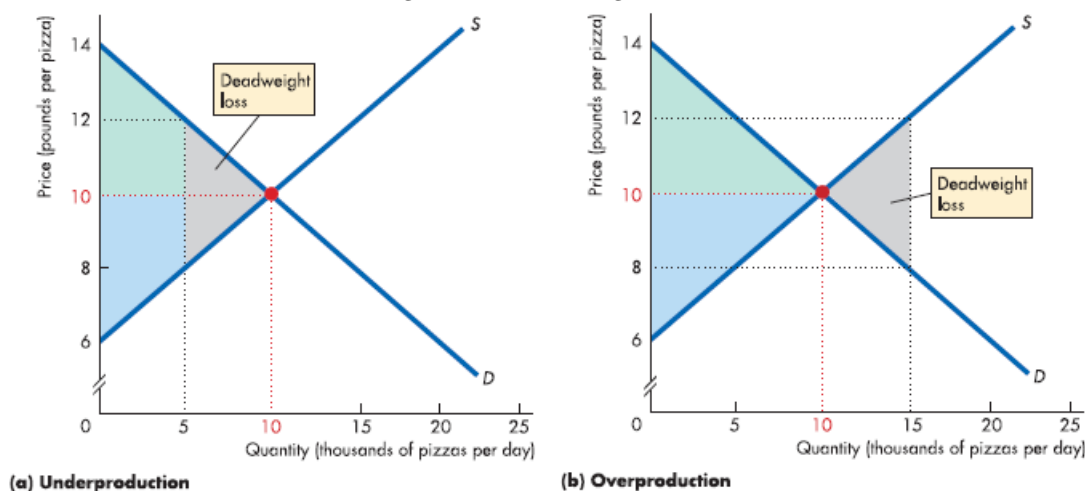
Pollution is a classic example of markets failing to reach allocative efficiency. Can you come up with an example of a case of market failure which leads to underproduction? (Hint: underproduction means we want more of the good or service to be produced – that must mean that all its ‘benefits’ are not being accounted for in our decision-making process.)

Key causes of market failures are³:

³Government intervention in a free market is one of the major causes of market failure in classical economics. The explanation for this is simple. When market failure has not occurred, we are operating at the most efficient point and if the government tries to interfere in the market (by lower prices, increasing wage, et cetera), they will move the market away from the efficient point. From this argument, we can conclude that governments should only interfere in markets in cases of existing or expected market failures.

We will discuss government’s interference in markets and its consequences in chapter 6, which is not a part of

Figure 5.3: Deadweight Loss



1. **Externality:** An externality is an effect on a third-party (can be individuals, a group of people, or society as a whole) that is caused by the consumption or production of a good or service. There can be both positive-externalities and negative-externalities. The reason externalities lead to market failure is because they are not usually considered in an individual's decision-making process⁴. As a result, the cost or benefit is under or overestimated, making us deviate from the socially desirable output level.

The pollution example is a case of negative-externality not considered in the production process, leading to overproduction. A single producer will never consider pollution as a part of his marginal cost. However, as a society, we want this activity to be restricted. The social equilibrium, in this case, is lower than the private equilibrium – this is a case of market failure since there is overproduction.

Second-hand smoking is another good example. A smoker may be aware of the dangers of smoking and consider that cost into his decision to continue smoking. However, if he is smoking in a public place, he must also consider the harm to others. Otherwise, we are not considering all the costs of this action, which will lead to overproduction (in this case – over-consumption of cigarettes in public places).

Planting trees in your garden is an example of an action that leads to positive-externality. Trees do not just benefit you and your garden – it benefits everyone in the surrounding area. If we do not consider this benefit to others in our decision-making, there will be underproduction.

your midterm syllabus. For now, in chapter 5, let us ignore governments (as we have done so far in this course), and discuss the other important causes of market failure. Once we have identified these causes, we can go on to discuss under what circumstances a government should and should not interfere in a market.

⁴We are usually only focused on the costs and benefits to ourselves only, not the wider world.

In each of these cases, we see that free-market is not taking us to the best possible outcome, indicating that market is not operating efficiently. Production is either more or less than what society as a whole will prefer. The solution to this is government, who can enact various rules and regulations to restrict and discourage activities that lead to negative externalities (ban on smoking in public places, tax on pollution, et cetera) and encourage activities that lead to positive externalities (developing free public parks, et cetera.).

2. **Common resource:** Whenever a group of people uses a shared resource (or, a resource lacks proper ownership), there will be overconsumption. This is something known as the **tragedy of the commons**.⁵ Without price rationing, individuals consume as much as they want, leading to the rapid depletion of the resource in question.

A group of experienced fishermen may all be individually aware of the fact that over-fishing now will lead to a depletion of fish-stock in the near future, leading to hardship for them and their families. However, if they do not trust each member of the group to reduce how much they fish, there will be a tendency for everyone to overfish simply because they do not want to be the only one to suffer.

Traffic-congestions are another very good example of the tragedy of the commons. Roads are a common resource used by everyone. Not surprisingly, there is overconsumption, leading to spiraling gridlocks. We all know that evenings are when most cars are out on the streets and if we go out as well, we will be worsening the traffic situation even more. If half of us agreed to stay home for Saturday-Monday-Wednesday evenings, and the other half agreed to stay home for Sunday-Tuesday-Thursday evenings, traffic-congestion in Dhaka city would all but disappear. But such a scheme will not work because we do not trust other people to keep up their end of the bargain and not use the roads on their designated stay-at-home days. Since there is no mutual trust, we will use the roads whenever we want to, instead of rationing our use of roads.

A government can, once again, impose these rules and see that they are properly followed.

3. **Public goods:** Public goods are goods that do not see a rise in cost of production due to a rise in number of consumers. For example, street-lights are public goods. Once they are produced and installed, the cost does not go up if the number of people benefitting from it goes up. National defense is another example – once this service is provided, everyone in the country is protected equally.⁶

Public goods lead to the **free-rider problem**. Once this service is provided, there is no motivation for anyone to pay for it, neither is there any means of making people pay for it. For example, it is not possible for the army to provide protection to your neighbor, who has paid the monthly fee, and exclude you from the service because you have not paid.

⁵If you end up doing a course of Environmental Economics of Resource Management, you will study this fascinating topic in greater details.

⁶Public goods have two defining characteristics: they are *non-rivalrous* and *non-excludable*. Non-rivalrous means that one person's consumption of the good will not diminish another person's consumption of it. Non-excludable means that once the service is provided, a person or a group of people cannot be excluded from enjoying it. But this is only a simplified understanding of public goods. If you are an Economics major, you may come across a higher course, usually 300 or 400-level, that solely focuses on Public Goods.

Once the borders are secured by the army, everyone (including your neighbor and you) in the country are equally safe.

Because of these factors, it is not possible to make a profit by producing public goods. As a result, there will usually be an underproduction of public goods since there is no motivation for the individual producers and sellers to supply the socially desirable level to the market, leading to market failure.

A government usually provides the public goods and public services to the citizens of the country.

4. **Monopoly:** To say a market is a monopoly does not necessarily mean that there is only one firm in the market. It simply means that a firm in that market has a large enough market-share to influence the quantity sold and the price-level.

A vegetable-vendor has no market power since if he tries to increase prices on his own, customers will just go to another vendor and buy the vegetables at a cheaper price. However, GrameenPhone has a monopoly power in the telecom industry of Bangladesh (even though they have other large and powerful competitors). If all their services were to become 20% more expensive tonight, tomorrow morning, we will continue to use their network.

A firm with monopoly power usually restricts production⁷ (underproduction) to drive up the price and maximize their profit, which, obviously, is not the socially desirable level.

5. **Other factors:** Asymmetric information, high transaction cost, et cetera. Refer to different books for other factors. In each of these cases, markets fail – production or consumption are either more or less than the socially desirable level, leading to a failure to maximize total surplus.⁸ In these instances, government interventions can rectify the situation through impositions of restrictions or incentives for the producers and consumers. That is what we will turn our attention to next – the manners in which governments interfere in failing markets and the outcomes of such interferences.

Let's sum up what we have learned in this chapter so far. I posed the question – do markets give us the most efficient outcome or can we do better than market equilibrium. By analysing surpluses, we saw that markets do indeed maximize welfare and any other outcome is sub-optimum.

However, then I introduced the concept of market equilibrium and saw that the market outcome⁹ need not always coincide with the socially desirable outcome. By this, I mean that individual producers and consumers will not always consider the effects of their actions on the wider society. And these leads to market failure since market has failed to give us the socially optimum outcome.

We made the argument that when markets fail, there is a valid argument for government intervention in the market.

In the next section, we are going to take a look at some case studies of government intervention in the market and their outcomes.

⁷Few weeks from now, we will be studying this phenomenon in details and try to figure out by exactly how much a monopolist will reduce output by and why. That will probably be our last topic before the Final Exam.

⁸It is possible that either the producer surplus or the consumer surplus is higher than their free-market equilibrium sizes. For example, under a monopolist, the producer surplus is higher than under scenarios without a monopolist. However, the total surplus will always be lower in cases of market failure.

⁹outcome of the result of individual decision makers

5.4 Government Interventions in Markets

There are four fundamental ways in which a government can influence the market. They can either try to keep price above or below a certain limit, or they can try to keep the quantity traded in the market above or below a certain limit. There are different ways to achieve each of these goals and in a very real sense, every government policy is meant to achieve at least one of these goals.

For the rest of this chapter, we are going to take a look at some case studies of government interventions in different markets.

5.4.1 Housing Markets and Rent Ceilings

Housing cost often represents the largest part of a person or a family's expenditure. It is not uncommon for a person or a family to spend upward of 50% of their total budget behind rent, mortgage repayment, or some other form of housing payment. Cities such as New York, London, or Tokyo are infamous for their high rents, with a monthly rent of a single-room reaching US\$2,000 and more. This is far beyond the affordability of most people. Even in Dhaka, house rents in certain parts of the city may be considered astronomically high for regular working person.

So what can be done about this?

Should something be done about this?

Politicians often make lofty promises during their election campaigns. Some of these promises may be false, but at least some are genuine. After all, any politician interesting in getting reelected has to fulfill at least some of his campaign promises. One of the most common promises made by politicians all over the world is a reduction in the costs of living. *Rent ceilings* is a common policy implemented by perhaps well-to-do, but also probably misguided, politicians. And on paper, it sounds like a good thing.¹⁰

Rent ceiling is an example of a price ceiling, when price is capped at how high it can rise. If this restriction is binding, it prevents price adjustment, leading to shortage.

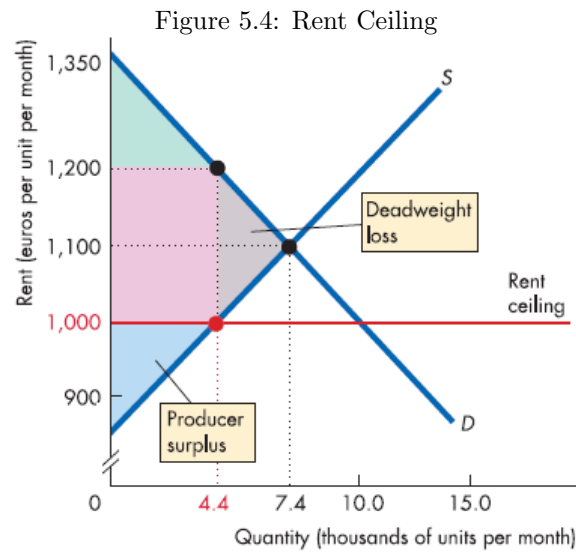
Like any other good or service, the house-market is regulated by demand and supply. Existing and interested tenants demand houses (often elastically – can you say why?) and landlords supply houses (often inelastically – can you say why?). These demand and supply interacts in the market to give us equilibrium price and quantity. By price, we mean rent, and by quantity, we mean the number of houses available for rent.

What effect will a rent ceiling have on this market? The same effect we see in any other market when the price of a good is set at a level beneath its equilibrium price – there will be excess demand (or, supply shortage) in the market. Remember our discussion about price adjustment? Refer back to figure 3.4 in page 24 if you do not.

Refer to figure 5.4. The free market equilibrium rent is 1,100, in which case, 7,400 units are demanded. However, the government has imposed a rent ceiling of 1,000 meaning that rent cannot be higher. Why did they do it? Because they wanted to lower the rent and make it affordable for people. But what is the outcome? A housing shortage. (Notice that if the government set a rent ceiling above the market price, it has no effect since the ceiling is not binding. A rent ceiling of 1,500 will not have any effect on the market.)

At this low rent, only 4,400 units are available for rent, against a demand of 10,000. So there is a shortage of houses in the market of 5,600 units. In their misguided benevolence, the government has ended up costing 3,000 families their homes. But that is only half the story.

¹⁰Maybe not to an economist, but who cares about them.



A rent ceiling of €1,000 a month decreases the quantity of housing supplied to 4,400 units. Producer surplus shrinks to the blue triangle and a deadweight loss (the grey triangle) arises. If people use no resources in search activity, consumer surplus is the green triangle plus the red rectangle. But if people use resources in search activity equal to the amount they are willing to pay for available housing (red rectangle), the consumer surplus shrinks to the green triangle.

Because now, there is only 4,400 houses available that has to be somehow allocated to 10,000 people. And this is where some major issues crop up.

- Search Activity

Whenever we engage in a transaction, we spend some of our time in **search activity**. We spend a few minutes asking friends and reading reviews online before deciding where to go eat. We visit different stores and try out different outfits before deciding which one to buy.

When houses are limited, people have to spend a considerably larger portion of their times trying to find the house. Time is a valuable and limited resources – time is money. The longer you have to spend in search activity, the more expensive the activity becomes. Initially, you may have been willing to pay 1,200 for a house. Because of the rent ceiling, you need to pay more than 1,000. However, the increased search activity is worth an additional 200. Search activity eats into your surplus.

- Black Market

How often have you wanted to go to a cricket match or a concert and found out that all tickets have been sold out? You go online, scroll through your news feed, and find a person who has an 'extra' ticket that he is willing to sell, but for taka 2,000 instead of taka 1,500, which is the original price.

This, of course, is illegal. You are not allowed to resell tickets. Certainly not at a higher price. But people are willing to pay a higher price. And so there will always be people looking to take advantage of that and make a nice profit.

Introduction of rent ceilings leads to landlords coming up with ingenious ways to earn extra money on the side. Is there an attached garage with the house? That will be an extra taka 3,000. Elevator? Taka 2,500 a month. Want to use the elevator after 10 p.m.? That will be taka 100 per use. Is it Eid? Pay a 10% bonus on top of your rent. A common charge imposed on new tenants is something called a 'key charge' or 'key money'. The new sets of lock and keys that have to be installed before you move in may end up costing you taka 10,000. You know that the landlord is overcharging, but what can you do? There are 5,600 people who are desperate to find a new house. If you try to argue with the landlord, he can give the lease to someone else instead of you. He has done nothing illegal after all. He has lowered the rent as the law required him to do.

Effectively, the rent ceiling is not really helping the tenants very much. There is an excess demand in the market of 5,600. It will be very easy for the landlord to find another tenant, but it will not be easy for the tenant to find another house.

The lesson here is that stopping rents from adjusting to the equilibrium level does not reduce scarcity¹¹. In fact, rent ceilings increase scarcity by reducing quantity supplied and increasing quantity demanded. A better solution will be to increase the supply of houses. A rightward shift of the supply curve would lead to a fall in rent without creating a shortage in the market. But, that is a long-term (and expensive) solution (and the temptation of a quick-fix often trumps basic logic).

5.4.2 Labor Markets and Minimum Wages

So we have seen that setting a bar on how high price can rise is not a smart thing to do. How about setting a bar on how low prices can fall? Let us look at the market for workers. Individuals offer their services to the market (labor supply) and firms hire these workers (labor demand). The interaction of demand and supply gives us the equilibrium price and quantity. By price, we mean the wage rate, and by quantity, we mean how many people are hired at the market wage rate.

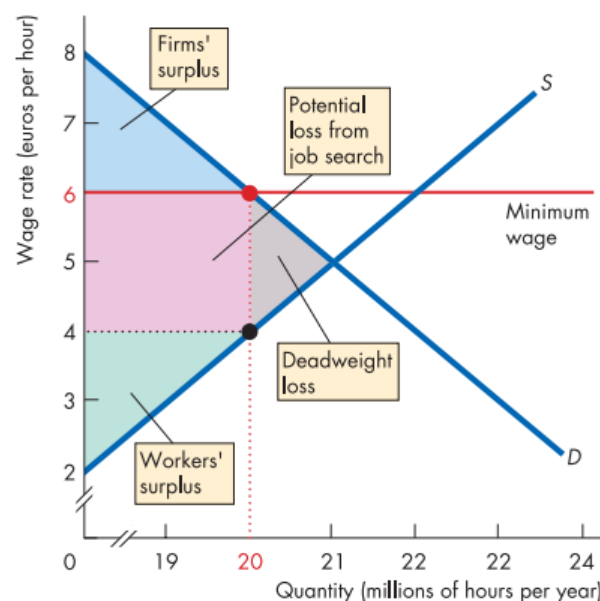
When government fixes a minimum wage level above the market wage rate, they create unemployment. Due to the high wage rates, employers immediately start hiring workers for fewer hours while at the same time, the high wage rate brings more workers into the market. As you can see in figure 5.5, the market equilibrium level is 21 million hours of work at a wage rate of 5. However, when the minimum wage rate is set at 6, quantity demanded falls to 20 million, quantity supplied increases to 22 million, and there is an unemployment in the market for 1 million hours of work.

With additional workers in the market, employers want to hire the best workers available. So they increase the *job requirements*. You suddenly need better degrees and more work experience and more impressive references to land the same job. A person who has worked at the same

Minimum wages are examples of price floor, where we restrict how low prices can fall. If minimum wage is binding, it prevents wage/price adjustment leading to unemployment.

¹¹which is the main reason for high rents

Figure 5.5: Minimum Wage



A minimum wage shrinks the firms' surplus (blue) and workers' surplus (green) and creates a deadweight loss (grey). If people use resources in job search equal to the amount they are able to gain by finding a job, the red rectangle is also lost.

factory for 20-years suddenly sees that he has to spend time and money to go get a 3-month training program or he risks losing his job. A fresh graduate sees his chances of landing a good job evaporate right in front of his eyes. A person who has taken a huge loan to get a degree with the hope of a good job comes to the depressing realization that he needs to take out more loans and get yet another degree or he will not get a good enough job. All these because we are now forcing the employers to pay the workers a wage rate above the market rate.

In addition, the workers who manage to hold on to their employments are under additional duress. They know that there are people out there, waiting for him to make one mistake, just one mistake, and the boss can fire him and hire someone else. After all, the employers now have to pay the workers a higher wage. Why should they not expect better performance?

Used to working from 9 a.m. to 5 p.m. usually? You can forget all about that now. There are people in that sea of unemployed workers who will offer to stay back until 7 p.m. to get some extra work done. So you better stay until 8 p.m. and keep your employer happy. Or you are out on the streets. Want to take a day-off because you are sick? No you won't. And remember that your office used to pay for lunch for all workers? Not anymore; they are paying you more, you can buy your own lunch. All your work benefits are going away.

Once again, a benevolent policy, aimed at helping workers live better lives, have back-fired and ended up hurting them. What would have been a better solution to this problem? One that takes time and money. A better educated, better trained, and healthier labor-body would

lead to a rise in demand. When the demand curve shift to the right, wage rate would go up but without creating unemployment. But who has time to educate a child and ensure he gets proper nutrition and wait 2-decades for him to grow up and enter the job market?

5.4.3 Taxation

In essence, tax is nothing but government revenue. If you consume that good (the government may say to buyers), or if you sell that other good (the government may say to sellers), or if you make this much earnings (the government may say to income-earners), or if you live in this part of the city (the government may say to citizens), or if you engage in any one of a hundred different things that I will keep an eye on, you have to give me this much money.

Some people say taxation is an essential component of a functioning, thriving society. Some people say taxation is nothing but institutionalized extortion. But there is no denying that in life, only two things are for certain: death and tax.

In a free market without government interventions, the amount that consumers pay for a good or a service and the amount received by the sellers is same. Taxes drive a wedge between this – consumers pay more, producers receive less, and the difference between these two amounts is the total **tax incidence**.

Tax incidence is the division of the burden of tax between buyers and sellers.

However, and here is where it gets interesting: even though governments can impose taxes, they cannot decide who pays the tax – that is decided by the market. Governments may impose a production tax (tax on sellers) or a sales tax (tax of buyers). However, the tax incidence, who pays how much of the tax, will be exactly the same in both cases.

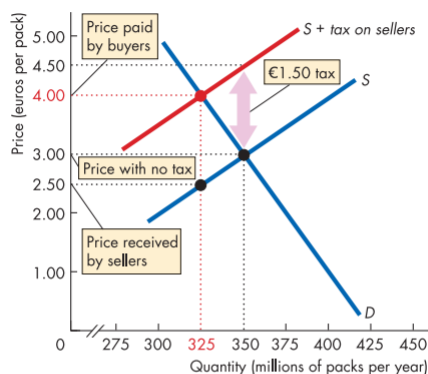
- A tax on sellers is like an increase in cost. It decreases supply. We add the tax amount to the minimum price that sellers are willing to accept (remember, the supply curve is the minimum-price curve), and that gives us the $S + tax$ curve.
- A tax on buyers lowers the amount they are willing to pay, which decreases demand. We subtract the tax amount from the maximum price that buyers are willing to pay (remember, the demand curve is the marginal willingness to pay curve) and that gives us the $D - tax$ curve.

Refer to figure 5.6. Tax incidence in each case is the same: \$1.0 on the buyers (66.67%) and \$0.5 on the sellers (33.33%). How much is government's total **tax revenue**? They receive $\$(4.0 - 3.0) \times 325 \text{ million} = \$325,000,000$ from the consumers and $\$(3.0 - 2.5) \times 325 \text{ million} = \$162,500,000$ from the sellers, for a total of \$487,500,000.

Draw a separate diagram with a demand and supply curve and label the equilibrium price and quantity properly. Introduce a tax (on sellers or buyers) in this market. Identify the areas that shows you the tax incidence on consumers and producers. Figure out what happens to the consumer surplus and the producer surplus. Identify the area that represents government revenue. Then identify deadweight loss. How would the tax incidences change if you changed the elasticity of demand/supply?

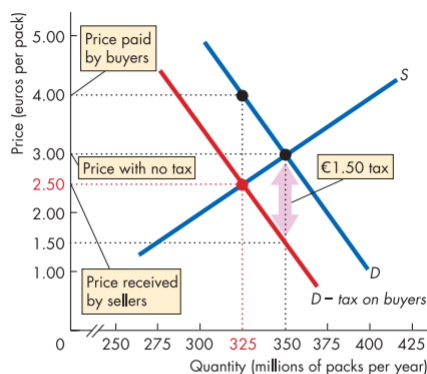
Figure 5.6: Tax Incidence

A Tax on Sellers



With no tax, 350 million packs a year are bought and sold at €3 a pack. A tax on sellers of €1.50 a pack shifts the supply curve leftward to $S + \text{tax on sellers}$. The equilibrium quantity decreases to 325 million packs a year, the price paid by buyers rises to €4 a pack, and the price received by sellers falls to €2.50 a pack. The tax raises the price paid by buyers by less than the tax and lowers the price received by sellers, so buyers and sellers share the burden of the tax.

A Tax on Buyers



With no tax, 350 million packs a year are bought and sold at €3 a pack. A tax on buyers of €1.50 a pack shifts the demand curve leftward to $D - \text{tax on buyers}$. The equilibrium quantity decreases to 325 million packs a year, the price paid by buyers rises to €4 a pack and the price received by sellers falls to €2.50 a pack. The tax raises the price paid by buyers by less than the tax and lowers the price received by sellers, so buyers and sellers share the burden of the tax.

A simple diagram that we discuss in class can have many different variants, depending on the parameter values. For example, elasticity will play a significant role in deciding who bears how much of the tax burden. Other factors, such as the type and size of tax imposed will factor into the final outcome as well. We will not go through all unique cases for two reasons: (i) I will not have enough time to cover everything without overburdening you with too much information, and (ii) if I do everything, it discourages you from trying to figure these things out on your own which you absolutely should be doing.

As a fun exercise, try to figure out on your own what happens when government imposes a tax on the following: (i) Perfectly Elastic Demand, (ii) Perfectly Inelastic Demand, (iii) Perfectly Elastic Supply, and Perfectly Inelastic Supply.

The important thing to keep in mind is that burden of tax is determined by market forces. So when, at the height of the nationwide student protest of 2015 about government's decision to impose VAT on education, our Finance Minister goes in front of the international media and says that the VAT imposed is on institutions (schools, universities, et cetera.) and students would not have to pay more, he harms his own credibility.

Demand of education, at any level, is inelastic in Bangladesh. The bulk of the incidence of any tax on education, therefore, would eventually get pushed onto the students.

What can the government do about this? Nothing. They will collect the VAT from institutions, sure. But that does not mean that if initially tuition was 100 – students were paying 100 and institutions were receiving 100 – tuition will remain at 100 but institutions will only keep 90 and pay 10 as tax. What would happen is that tuition would go up to 110. Students would pay 110 as tuition (but no tax! Yay!) and institutions would only keep 100 and pay 10 as tax.

You, as a student of ECO101, understand this market phenomenon now. So did the protesting students. They carried on until the VAT was repealed. Passion and some indignation are important in protests, but you also have to keep your wits about you. So kudos to the students. Let's hope that they carried some of that passion and zeal with them into the classrooms when they returned.

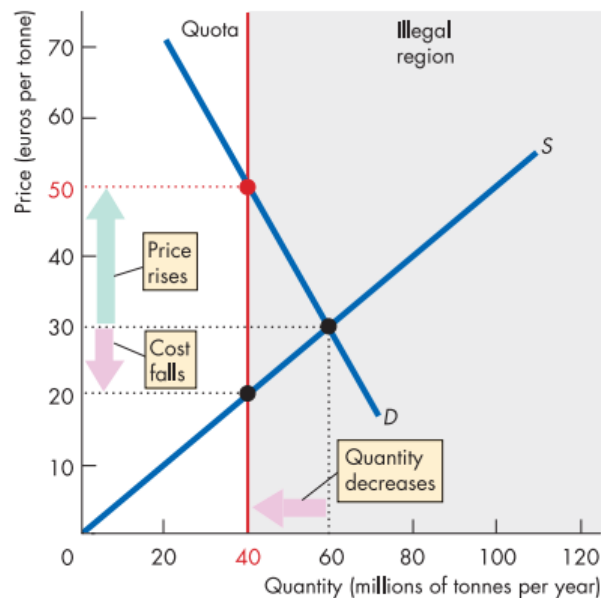
5.4.4 Different types of Support for Producers

Now let us look at three separate schemes that governments try to implement to help out producers (almost always farmers, producers of agricultural products): (i) production quotas, (ii) subsidies, and (iii) price supports.

Production quota is the maximum amount or quantity of a good that can be produced within a given period of time.

Production Quotas: A production quota puts an upper limit on how much of a good can be produced. This lowers the quantity supplied to the market, thereby increasing the price level. This can be beneficial to producers of goods with inelastic demand, such as rice in Bangladesh. With an inelastic PED and increasing price, farmers enjoy higher revenue.

Figure 5.7: Production Quota



With no quota, 60 million tonnes a year are produced at €30 a tonne. A quota of 40 million tonnes a year restricts total production to that amount. The equilibrium quantity decreases to 40 million tonnes a year, the price rises to €40 a tonne, and the farmers' marginal cost falls to €20 a tonne. In the new equilibrium, marginal cost (on the supply curve) is less than marginal benefit (on the demand curve) and a deadweight loss arises from underproduction.

Refer to figure 5.7. The entire gray-area is illegal. Production cannot exceed 40 million tonnes. However, how do you implement this rule? There are farmers spread all over the country and it is a very intensive and expensive activity to keep an eye on every single farmer. The government's best bet is to assume that farmers will follow the honor-code and not over-produce. However, with the high price, farmers want to supply 100 million tonnes to the market.

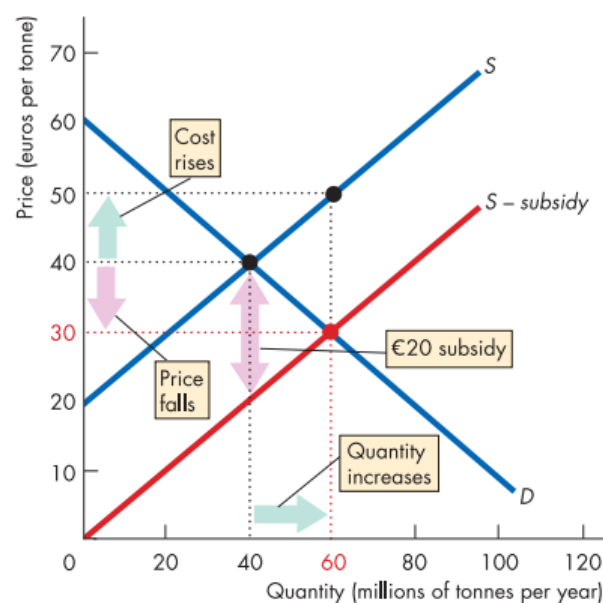
Every single farmer will be trying to produce more than the legal limit, arguing that if only

he produces more and enjoys extra profit, no one is harmed. Which is true. The problem is that if everyone starts to think that way and everyone over-produces, the schemes fall apart. There will be over-production (higher than the initial equilibrium quantity of 60 million tonnes) and farmers are now worse off than they were. Yet another government scheme to help people backfires.

Production Subsidies: A production subsidy lowers the cost of production and thereby increases quantity supplied and lowers price. However, government payment to producers often ends up being a substantial amount, there will be deadweight-loss in the market, and the production level can be deemed as an inefficient.

Subsidies are payments made by the government to producers for each unit produced.

Figure 5.8: Production Subsidy



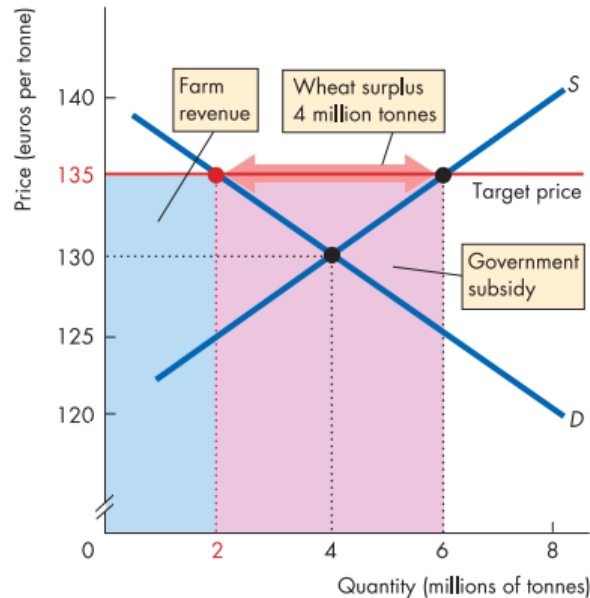
With no subsidy, 40 million tonnes a year are produced at €40 a tonne. A subsidy of €20 a tonne shifts the supply curve rightward to $S - \text{subsidy}$. The equilibrium quantity increases to 60 million tonnes a year, the price falls to €30 a tonne, and the price plus subsidy received by farmers rises to €50 a tonne. In the new equilibrium, marginal cost (on the blue supply curve) exceeds marginal benefit (on the demand curve) and a deadweight loss arises from overproduction.

Refer to figure 5.8. Subsidy lowers marginal cost of producers, leading to quantity increasing to 60 million tonnes and price falling to 30. This overproduction distorts the market and introduces deadweight-loss. Government also has to pay a huge sum as subsidy to producers (euro 1.2 billion).

A price support is a government-guaranteed minimum price for a good.

Price Support: Price support schemes are only binding if they are set above the equilibrium price (if the guaranteed price by the government is more than the market price). This lowers quantity demanded and increases quantity supplied, forcing the government to buy up all the extra surplus production at the higher guaranteed price.

Figure 5.9: Price Support



With no price support, 4 million tonnes are produced at €130 a tonne. A price support of €135 a tonne raises the price to €135 a tonne, decreases the quantity sold to 2 million tonnes, and increases the quantity produced to 6 million tonnes. The price support creates a surplus of 4 million tonnes. To maintain the price support, the government buys the surplus for €540 million a year. If the government does not buy the surplus, the price returns to €130 a tonne.

Refer to figure 5.9. Market price level is 130. However, the government guarantees a minimum price of 135 to the farmers, meaning that if farmers cannot sell all their products at the market for at least 135, government will buy up all the unsold products. What this does is increases production from 4 million tonnes to 6 million tonnes and farmers are unwilling to sell them for anything less than 135. This drops the quantity demanded for 2 million tonnes.

As a result, the government has to buy 4 million tonnes of unwanted output from the farmers and then pay even more money in storing these mountains of foods in the warehouse. But no one wants these expensive produce. Eventually, the government has to spend even more money to destroy these expensive, unwanted products.

About 49% of all food we produce globally is wasted. And schemes such as these play an important part. Farmers' Associations in the Euro zone are very strong and very politically

connected. EU pays more than euro 200 billion annually¹² in buying over-priced products from farmers that no one wants, storing them in warehouses, and then destroying them when they begin to spoil. But this is an example of politics in action, not economics.

¹²EU countries have been known to spend more than 50% of its budget behind these wasteful price support schemes.

Chapter 6

Output, Cost, and Competition

In this chapter, we turn our attention to producers. Namely, producers' decisions about how much to produce, and how to produce.¹

As a consumer, you are able to make decisions on the go. When you are hungry, you go to a restaurant of your choosing and order a plate of food. If that is enough, you pay and leave. If not, you order a second plate. The only thing you had to plan for is make sure you have enough money and time to consume two (or however many) plates of food.

However, consider the producer's (in this case, the restaurateur's) decision. The night before you walked in and placed your order, the chef would have had to anticipate the customer-volume for the next day and prepare the meals or at least have enough ingredients available. Before that, a decision has had to have been made about what size of pots and pans and ovens to have in the kitchen. Before that – the location of the restaurant, number of chairs and tables for the customers, the type of cuisines to serve – all these decisions have been made months and maybe even years before the restaurant even opened its doors to customers.

A producer is often not able to instantaneously make decisions as we consumers are. We will soon see that that is because producers have to deal with both fixed factors and variable factors of production. Before we discuss what we mean by fixed and variable, let us define two different types of time-frames that producers face in decision making.

In the short-run,
quantity of at least
one factor of
production is fixed.

In the long-run,
quantities of all
factors of production
can be varied.

Short-run decisions are easily made and easily reversed. Long-run decisions, not so much. For example, hiring day-laborers to work in your factory is a short-run decision. You may hire him today, but not again tomorrow. Easily made and reversed. However, when you hand out a contract to an employee, that decision is not easily reversed. You have to continue paying him until his contract duration runs out. In the short-run, hiring workers is a changeable factor of production, but hiring a factory manager is a fixed factor of production. In the long-run, when the manager's contract expires, you can decide whether you want to renew the contract or get rid of the manager. In the long-run, all factors of production are changeable.

In a nutshell, within a given period of time, if you are able to change the quantity of all your factors of production, you are operating in the **long-run**. If not, and you find that at least one of your factors of production is fixed, then you are operating in the **short-run**. In the short-run, if a firm wants to increase its production, it will probably have to hire more workers. However, in the long-run, the firm can decide between hiring more workers and buy a new machinery that increases productivity.

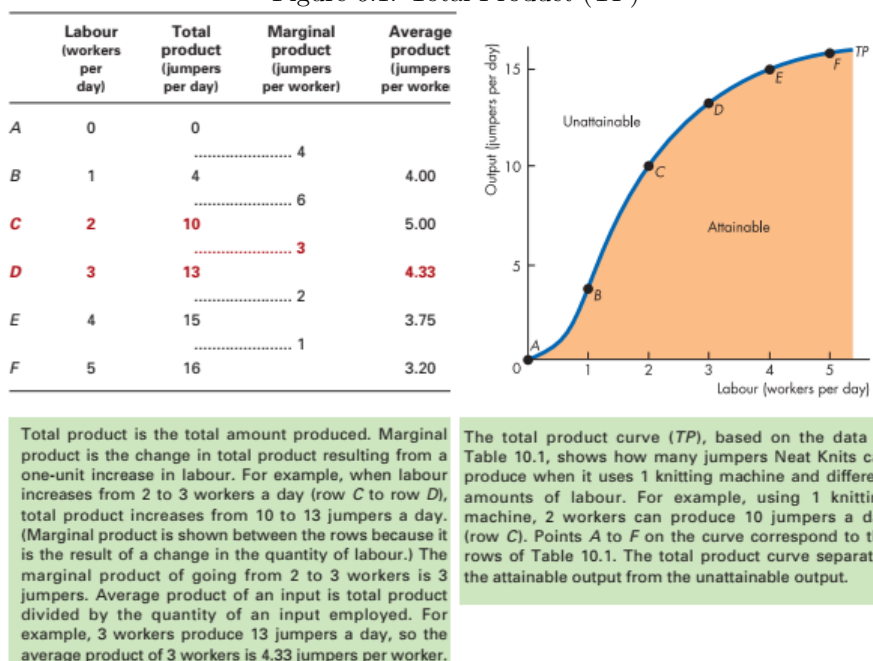
¹Ideally, we would have spent sometime looking at consumers' decision-making process as well, but we will leave that for another course.

6.1 Technology and Cost

6.1.1 Short-run Technology Constraints and Production

Total Product is the maximum output that a given quantity of labor (or any other variable inputs, such as raw material) can produce. **Marginal Product** of labor (or any other variable input) is the increase in total product resulting from a one-unit increase in the quantity of labor employed, with all other inputs remaining the same. **Average Product** is the productivity of workers on average.

Figure 6.1: Total Product (TP)



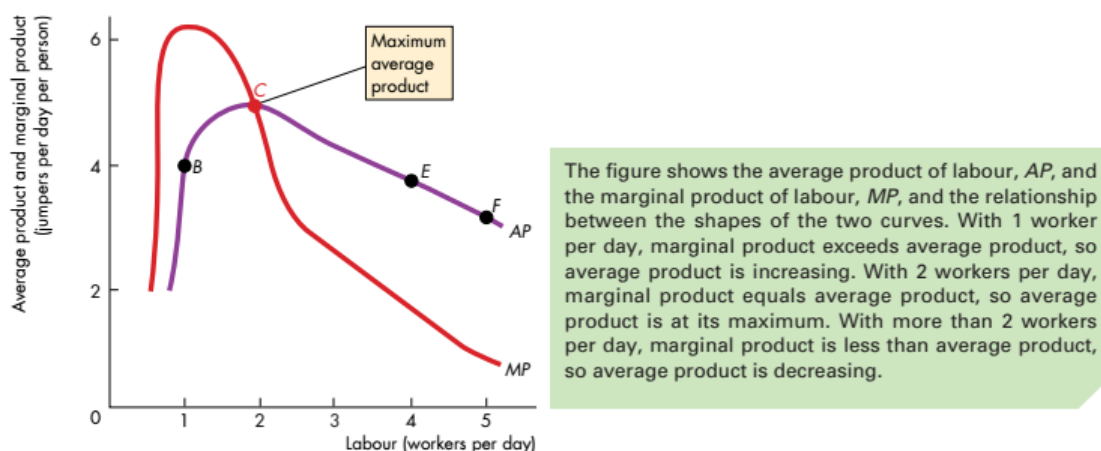
Refer to figure 6.1. As we hire more workers, output (total product) increases. However, it does not increase at a constant rate. At lower levels of production, firms enjoy **increasing marginal returns**. Each worker gives us more output than the previous worker. This is not to say that one worker is necessarily better or more skilled than another. What happens is that as we hire more workers, workers can **specialize**, which makes them more productive.

Adam Smith talked about this in the first pages of his seminal book, The Wealth of a Nation.

He visited a pin factory and observed that the pin-making process can be broken down into 18 distinct steps. Each worker, performing all these steps, could produce 10 to 20 pins a day. However, when workers specialize and focus on individual tasks, Smith observed in another factory, 10 workers were able to produce 48,000 pins per day!

This is one of the most famous examples of 'specialization' and 'division of labor' within the economics literature.

Figure 6.2: Average Product and Marginal Product



No let us take a look at figure 6.2. Both curves go up and then down – they have an inverted-U shape. If you understand the rationale behind the shape of a TP curve, then understanding MP becomes trivial. In the early stages of production, when a firm experiences increasing returns, hiring of each additional worker (the marginal worker) gives more output than the previous worker had (increasing MP). However, once the firm enters the diminishing return level of production, each new worker gives lower output than the previous worker (decreasing MP). That explains why the MP curve exhibits an inverted-U shape. Try to figure out, on your own, why the AP curve is shaped similarly. It should not be very difficult.

If the marginal value is above the current average, then the new average will be higher than the current average, vice versa. Or, if $M > A_{old}$, $A_{new} > A_{old}$

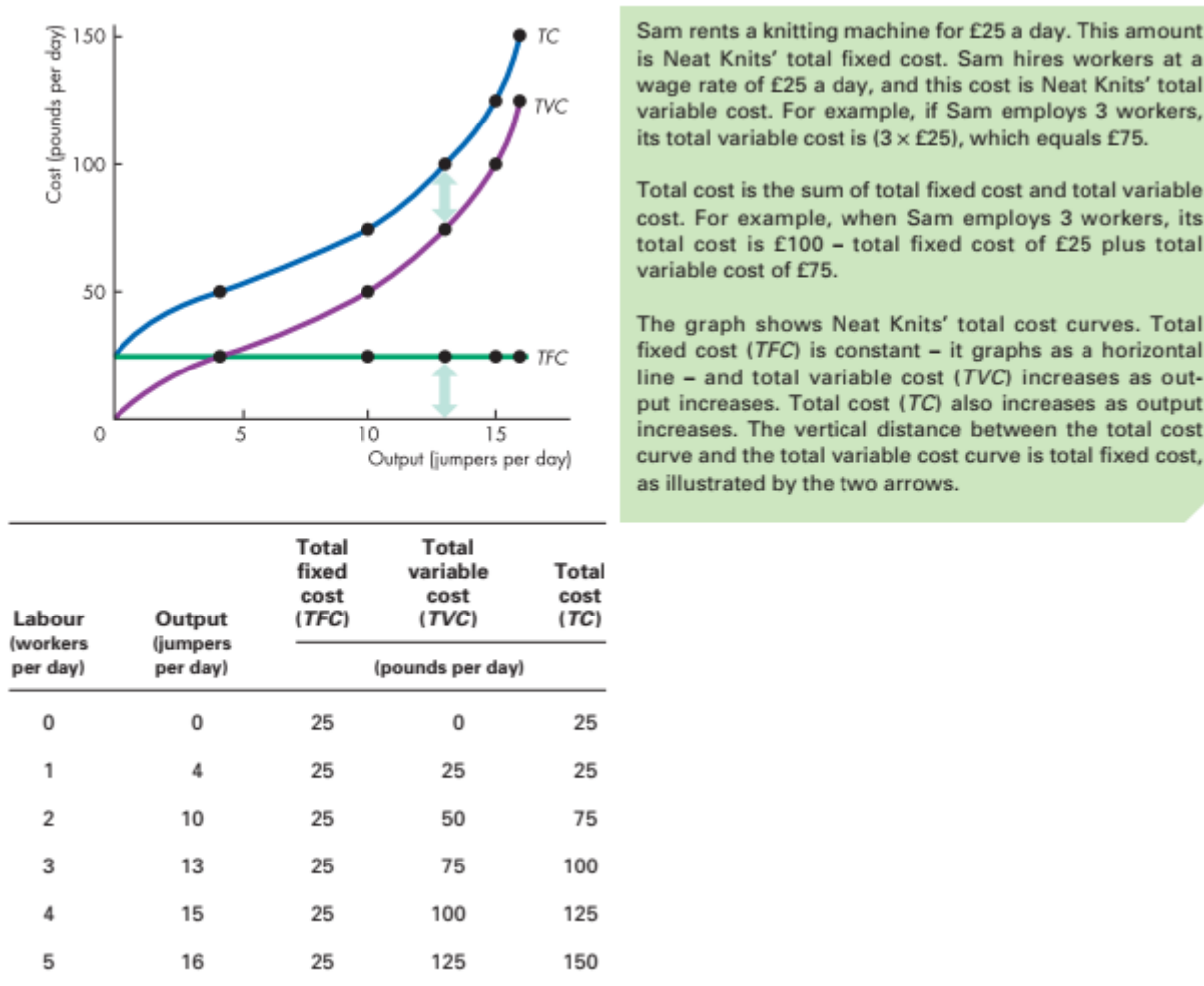
Relationship between Average and Marginal

Think of a numeric example if the relationship between margin and average is not clear to you. At this stage in the course, you have sat for 2 out of the 3 quizzes that will be taken. Suppose, your average score in the first 2 quizzes is 10. Your score in the 3rd quiz is your marginal score. If you get more than 10 in the 3rd quiz (your marginal score is higher than your current average), your new average will go up. If not, your average goes down.

This is why, initially, the TP curve is increasing at an increasing rate. Each new worker allows further division of labor, which increases the overall productivity. Once the full benefits of division of labor has been extracted, adding more workers lower productivity. Once again, this does not mean that each new worker is less skilled than the previously hired workers. It's just that the number of workers can no longer work together productively, given the fixed components. For example, each factory space will have an upper limit on how many workers can work there. Squeezing in more workers will result in them getting in each others ways and impeding smooth operation. Similarly, an equipment such as a hammer can only be used by one worker at a time. Anyone else in need of the hammer has to stand around idly, unproductively, waiting for the hammer to become free. This is why the latter stages of a TP curve is still increasing, but at a decreasing rate. This phenomenon is known as **diminishing marginal return**.

Marginal cost curve is simply the 'rate of change' of a TP curve. Therefore, MP curves

Figure 6.3: Short-run Costs



are inverted-U shaped - rising initially and then falling. Average productivity curve is similarly inverted-U shaped, and the MP curve intersects the AP curve at AP's maximum point.

6.1.2 Short-run Cost Structure

In the short-run, remember, a firm has both fixed and variable factors of production. **Total fixed costs** (TFC or FC) is the cost of a firm's fixed factors. This includes cost of rent, for example. Fixed costs do not change with production. **Total variable costs** (TVC or TC) is the cost of a firm's variable inputs. This includes wages, cost of raw material, et cetera. Variable costs increase as production increase. **Total cost** (TC) is the cost of all factors of production. Figure 6.3 plots the TVC, TFC, and TC of a firm's production.

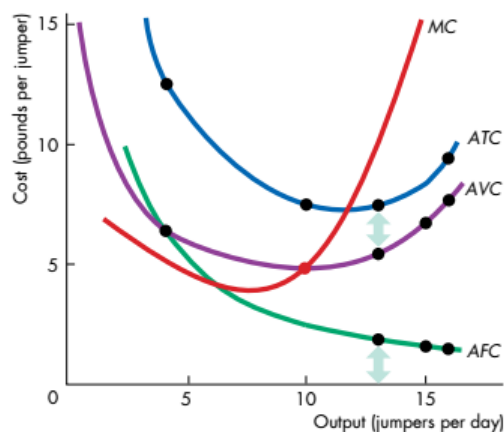
Marginal cost is the cost incurred from production of one additional unit. Marginal cost decreases at low outputs because of economies from greater specialisation. It eventually increases

$$TC = TFC + TVC$$

$$\text{Marginal cost} = \Delta TC / \Delta Q$$

because of the law of diminishing return. **Average fixed cost** (AFC) is total fixed cost per unit of output. **Average variable cost** (AVC) is total variable cost per unit of output. **Average total cost** (ATC) is total cost per unit of output.²

Figure 6.4: Average Cost (AC) and Marginal Cost (MC)



Marginal cost is calculated as the change in total cost divided by the change in output. When output increases from 4 to 10, an increase of 6, total cost increases by £25 and marginal cost is $\text{£}25 \div 6$, which equals £4.17. Each average cost concept is calculated by dividing the related total cost by output. When 10 jumpers are produced, AFC is $\text{£}25 \div 10$, AVC is $\text{£}50 \div 10$, and ATC is $\text{£}75 \div 10$.

The graph shows the marginal cost curve and the average cost curves. The marginal cost curve (MC) is U-shaped and intersects the average variable cost curve and the average total cost curve at their minimum points. Average fixed cost (AFC) decreases as output increases. The average total cost curve (ATC) and average variable cost curve (AVC) are U-shaped. The vertical distance between these two curves is equal to average fixed cost, as illustrated by the two arrows.

Labour (workers per day)	Output (jumpers per day)	Total fixed cost (TFC)	Total variable cost (TVC)	Total cost (TC)	Marginal cost (MC)	Average fixed cost (AFC)	Average variable cost (AVC)	Average total cost (ATC)
		(pounds per day)			(pounds per jumper)			
0	0	25		25				
					6.25			
1	4	25	25	50		6.25	6.25	12.50
					4.17			
2	10	25	50	75		2.50	5.00	7.50
					8.33			
3	13	25	75	100		1.92	5.77	7.69
					12.50			
4	15	25	100	125		1.67	6.00	8.33
					25.00			
5	16	25	125	150		1.56	7.81	9.38

Figure 6.4 plots the AVC, AFC, ATC, and MC curves for a firm. It is easy to plot them, once you have worked out the table beneath the graph. However, you should also intuitively understand why the curves are shaped the way they are.

In addition, I also expect you all to understand what factors will lead to shifts of which curves, even though we will not spend too long of our class time discussing them. For example, better technology will allow us to produce more at lower cost. Therefore, ATC and AVC should shift down, but AFC will remain constant. Rise in rent will shift AFC and ATC up, but not

²The marginal cost curve will always intersect the average variable cost curve and the average total cost curve at the minimum point. The cause behind this relationship is the same as when we discussed AP and MP.

Figure 6.5: Long-Run Production and Average Cost

Workers used	Number of Printouts (Per day)					
	Printer 1		Printer 2		Printer 3	
		AC1		AC2		AC3
1	1,000	1.00	1,500	1.33	2,500	1.20
2	1,750	0.80	2,500	0.96	3,500	0.97
3	2,500	0.72	3,250	0.86	4,250	0.89
4	3,000	0.73	3,350	0.96	4,750	0.88
5	3,250	0.80	3,500	1.03	5,000	0.92

AVC or MC. At this stage in the semester, you should be able to work these things out on your own.

6.1.3 Long-run Cost Structure

In the short-run, a firm is constrained by its fixed-factors of production. If it wants to increase production, it has to do so by increase the variables factors: hire more labor, use more raw materials, et cetera. However, it cannot alter its fixed factors: buy more machineries, rent a bigger factory, hire more foremans, et cetera. In the long-run, this is no longer an issue. Given a sufficiently long time period, a firm can change all its factors of productions – all costs are variable. A firm may, for example, choose to produce x units of a good, using 1 set of equipment, 2 set of equipment, or any number of equipment, whichever gives the lowest cost.

Suppose I run a printing business - a have a small shop on campus and I printout documents for students and the members of faculty. Cost of using a printer is 1,000 taka everyday. I do some analysis and find that I can print 1,000 pages per day working alone.³ The cost of hiring someone to help me in the store is 400 taka per day. If I hire someone to help me, we can print out 1,750 pages per per day. If I work alone but install a second printer, I can print out 1,500 pages per day. If I do both – get a help and a second printer – prints per day goes up to 2,500. The full table is given in table 6.5.

Average costs in each scenario is given in red-font. For example, when I use 2 printers and hire 3 workers, my total cost is $(2 \times 1,000 \text{ taka}) + (2 \times 400 \text{ taka}) = 2,800 \text{ taka}$. My average cost is $2,800 \text{ taka} / 3,250 = 0.86 \text{ taka}$.⁴

³Don't forget, the work is more than just clicking 'print'. You have to take instructions from the client, plug his USB-device into your computer, open the document, wait for print to complete, verify everything is in order, staple the document, and accept payment before you can move onto the next client.

⁴Remember, you are one of the workers. So when you have a total of 3 workers in the store, you are actually paying wages to only 2 of them.

Suppose my market research tells me that the demand per day will be 2,500 pages per day. Should I use 1 printer or 2 printers? Look at the table. Using 1 printer, I can print out 2,500 pages at an average cost of 0.72 taka per page. Using 2 printers, the average cost goes up to 0.96 taka per page. It is better for me to use only 1 printer. What if my business expands and now, the demand is 3,500 pages per day? Once again, look at the table. We cannot produce 3,500 pages with only 1 printer. But by using 2 printers, we can complete the task with an average cost of 1.03 taka per page. If we use 3 printers, the cost falls to 0.97 taka per page. Therefore, we will use 3 printers.

It should be intuitively obvious, therefore, that the short-run average cost analysis that we have done earlier in the chapter will not be appropriate for a long-run average cost analysis. This will become clearer once we draw the different average cost curves on a diagram and pull out the long-run average cost curve (LRAC) from them, which we have done in figure 6.6. I am using the graph used in the textbook. Feel free to draw the graphs of the printer-shop example in your own time. It will be a good practice.

Also, please be aware of the difference between drawing, sketching, and plotting a graph.

The decreasing part of LRAC curve exhibits **economies of scale** – as we produce more, our average cost goes down. The increasing part of LRAC curve exhibits **diseconomies of scale** – more production leads to a rise in average cost.⁵ If you do not know what they are, I have discussed them in class and the assigned textbook also explains what they are. However, you may also use this as a good opportunity to use your understanding of this chapter to try and figure out why LRAC may fall and why it may rise. Give it a go. Working out an answer on your own is often more rewarding than than asking someone about it.

6.2 Perfect Competition

This section will be the most critically challenging chapter of this course. However, if you manage to power through the complexities and come out on the other side with a proper understanding of the material, this chapter will also be the most interesting and rewarding chapter of this course.

6.2.1 What is Perfect Competition?

The word '*perfect*' has two meanings in english. We use them both in our everyday speech without noticing the subtle difference between them. The first meaning of '*perfect*' is the attribute of having all the required or desirable elements, qualities, or characteristics – simply, to be as good as it is possible to be. When we talk about *perfect competition*, we do not mean we have the best competition. The second meaning for the word '*perfect*' is the attribute of being absolute, or complete.

When you see a person and say, "*He is a perfect stranger to me.*", you do not mean to say that the person is the best stranger that you have ever met. You mean that he is a complete stranger to you! Similarly, when we say *perfect competition*, we mean to say that competition in the market is complete. Absolute.

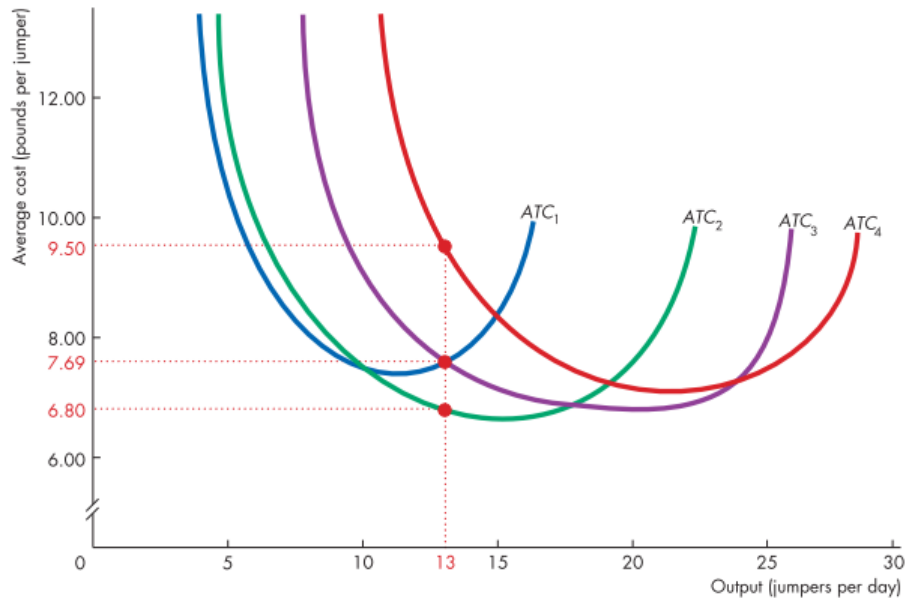
What does that mean?

It means that there are no impediments to competition. More precisely, when the following criteria are satisfied, we have perfect competition:

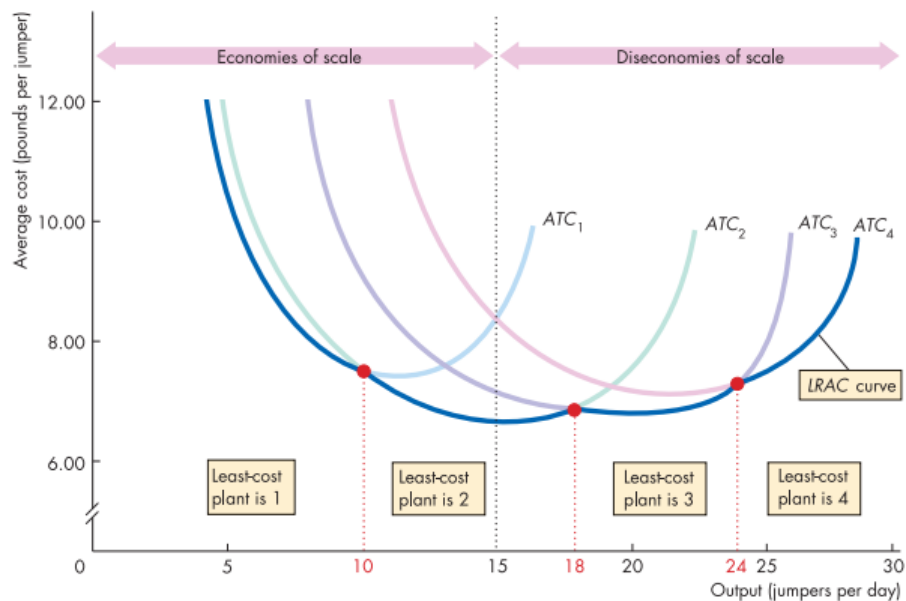
- Many firms, selling an identical product, to many buyers

⁵Production, of course, is a relative term. In the print-shop, production means number of pages printed. In a barbershop, production means number of haircuts provided.

Figure 6.6: Short-run AC curves and the Long-run AC curve (LRAC)



The figure shows short-run average total cost curves for four different quantities of capital. Neat Knits can produce 13 jumpers a day with 1 knitting machine on ATC_1 , or with 3 knitting machines on ATC_3 for an average cost of £7.69 per jumper. It can produce the same number of jumpers by using 2 knitting machines on ATC_2 for £6.80 per jumper or with 4 machines on ATC_4 for £9.50 per jumper. If Neat Knits produces 13 jumpers a day, the least-cost method of production – the long-run method – is with 2 machines on ATC_2 .



In the long run, Neat Knits can vary both capital and labour inputs. The long-run average cost curve traces the lowest attainable average total cost of production. Neat Knits produces on its long-run average cost curve if it uses 1 machine to produce up to 10 jumpers a day, 2 machines to produce between 10 and 18 jumpers a day, 3 machines to produce between 18 and 24 jumpers a day, and 4 machines to produce more than 24 jumpers a day. Within these ranges, Neat Knits varies its output by varying its labour input.

- No restrictions or barriers to entry of new firms or exit of existing firms
- Established firms enjoy no benefit over new firms
- All sellers and buyers are well informed about prices (there is no asymmetry of information)

It should be obvious why each of these is needed for competition in market to be perfect. Many firms and many buyers mean that no one buyer or seller has market power. If a seller tries to charge you too much (as is the case in monopolies), you can go to another seller who charges a fair amount. If a buyer wants to pay too little (as is the case in monopsonies), no one sells to him. Identical product, similarly, ensures that no one has any market power. The moment a firm starts to sell a differentiated product, he gains market power. We cannot have that in perfect competition. Similarly, if there were restrictions to entry of new firms, that would allow existing firms from enjoying high profit from preventing competition from entering the market. This once again would prevent perfect competition.⁶

Think of a vegetable vendor, selling vegetables from a cart. Vegetables sold on street is an example of a perfectly competitive market. Many sellers, selling the same vegetables, to many buyers.

If there was only one seller, he could charge a high price and only sell to the buyers willing and capable of paying the high price. If there was only one buyer, he could make the sellers compete against each other and buy from the lowest bidder. If everyone was selling local vegetables and one seller was selling broccoli, he would have market power. If none of these happens, we have a perfect competition.

No one can prevent a new vegetable seller from entering the market, for example. If a person thinks that there is money to be made from this endeavor, he is free to do so.

The takeaway from this, and an important fact to remember, is that both buyers and sellers are **price-takers** in a perfect competition. Price is fixed in the market through the interactions of hundreds and thousands of buyers and sellers, with no single agent capable of influencing this.

The next important fact to remember is that in a perfect competition, Price = Marginal Revenue. This relationship should be obvious. A firm cannot influence the market price level. Therefore, selling more does not mean price is falling. Therefore, whatever the price level in the market is, is also the firm's marginal revenue. Graphically, this is a horizontal line, indicating that price remains constant.

Remember these two relationships. We will return to them over and over again.

6.2.2 Revenue, Cost, and Economic Profit

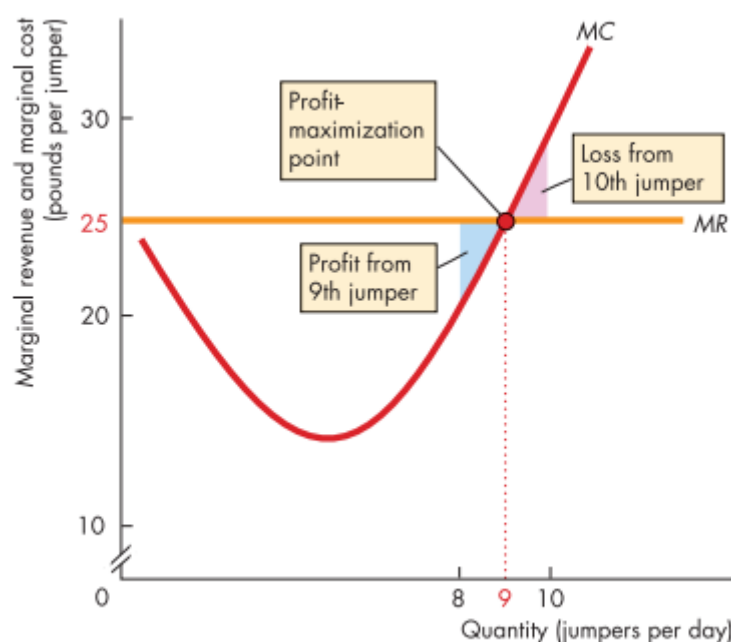
There are different ways of calculating the profit-maximizing output level of a firm. For example, Profit = Total Revenue - Total Cost. Total revenue of a firm in a perfect competition is a straight-line as we have discussed already. Therefore, the profit function is derived by simply multiplying the market price with quantity. If the total cost function is known, it is easy for us to calculate profit.⁷ Refer to the textbook for a worked-out numerical example. Pay close attention to the figures.

⁶We will discuss what these barriers to entry can be in the next chapter.

⁷It becomes simpler still if we know some basic calculus. Once we have the profit function, we only have to differentiate it, with respect to quantity, and set the first-order derivative equal to zero.

A price taker is a buyer or a seller who cannot influence the market price level, and accepts the price as is.

Figure 6.7: Profit-maximizing Output



We are going to find the profit-maximizing output level of a firm using an alternate method that may not be intuitively obvious straightaway. But once we understand the relationship, we will see that this method allows for much more versatile analyses to be conducted. We have already done a similar analysis before. During our discussion on social benefits and costs, we saw that by setting $MSB = MSC$, we could maximize social welfare. We called it *marginal analysis*. We are going to just that – do a marginal analysis – but this with, with marginal revenue and marginal cost.

In a nutshell, a firm maximizes profit when it produces a level of output which has the same level of MR and MC. This is illustrated in figure 6.7. This should make sense if you think about it a little. If we are to produce at a point where $MR > MC$, the cost of producing the next unit is lower than the price at which we can sell it in the market. So, we should produce more. Conversely, if we are producing at a point where $MR < MC$, we are incurring a higher cost by producing goods than the price we are receiving in the market. Therefore, we should produce less. Regardless of how much we are producing (and in whatever market structure we find ourselves in), profit-maximizing goals will mean we converge to the point where $MR = MC$.

We can think of this as a type of an equilibrium (but not really). Recall, once again, how we defined an equilibrium. An equilibrium was a point at which economic agents had no motivation to move away from, unless something external changed. At $MR = MC$, we have that. Producing more or less will mean a fall in profit. (Producing less would mean MR falls by a higher amount than MC and producing more would mean MC cost rise by a higher amount than MR.) Therefore, we settle at $MR = MC$.

But here is a very important thing to remember before we move onto the next section: profit maximization does not necessarily imply that the firm is making a positive-profit. If you find

yourself with an unfavorable cost structure or market conditions, it may not be possible for you to earn a positive profit. In that case, equating $MR=MC$ will still *maximize profit*, but that only means that you are minimizing your loss. This is the third important thing you should remember.

Different Profit and Loss Scenarios

In the short-run, a perfect competition firm can face one of these three situations:

1. **Economic profit**, meaning $price > ATC$

This is easy to understand. The price received in the market is higher than the firm's average cost of production. Therefore, the firm is making an economic profit. Try to draw a graph that represents this relationship.

2. **Economic loss with no production (temporary shutdown)**, meaning $ATC > AVC > price$

The market price, in this scenario, is lower than both the average cost and average variable cost of production. The fact that price is less than average variable cost is important as it implies that every time the firm produces one unit, it loses even more money. Therefore, the firm should stop producing.

However, recall that we are in the short-run now. There are factors of production, a 2-year lease agreement, for example, that cannot be thrown away. The agreement has to be honored. A bank loan that has to be repaid over a number of years is a similar example. As long as these fixed expenditures exist, the firm cannot just shutdown. Furthermore, in the long-run, the firm may expect something to change – a rise in price or a fall in its cost of production – that makes it profitable again. Therefore, in this scenario, the firm will only *temporarily* shutdown, produce nothing, and continue to pay its fixed costs. Try to draw a graph that represents this relationship.

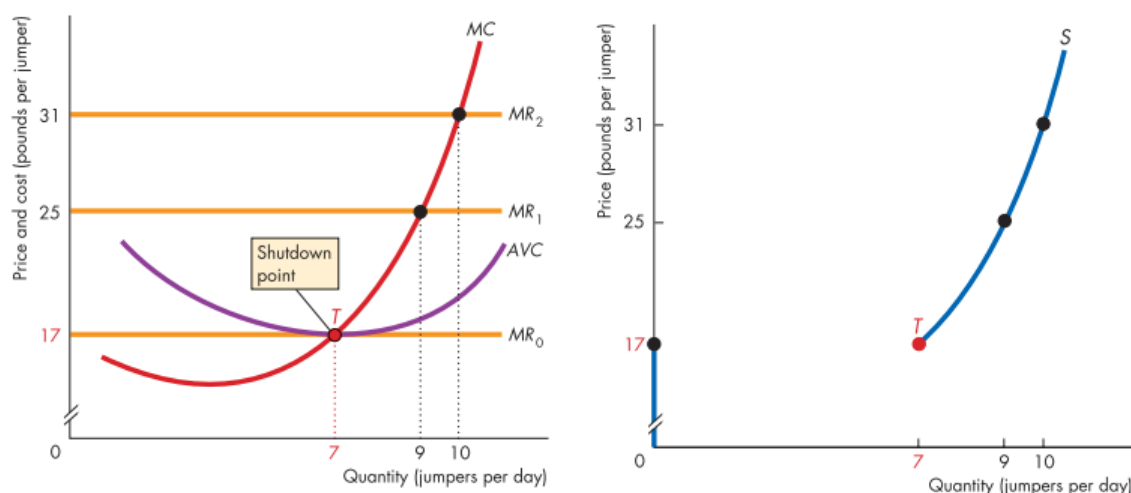
3. **Economic loss with production**, meaning $ATC > price > AVC$

In this case, the market price is lower than the firm's total cost of production, but higher than its average variable cost. The fixed costs, remember, are not directly related with production. They will remain constant regardless of whether we produce more or less or nothing at all. Ignoring the FCs for now, we see that if the firm continues production, its (variable) cost of production is lower than the money it earns in the market. The firm makes money if it continues production, even though it continues to make a loss.

For example, suppose that the firm has a fixed cost commitment of 500 taka every month. It can produce one unit of whatever it produces for 5 taka and sell it for 7 taka in the market. There is a market demand for 100 units. Therefore, if the firm produces, it makes a loss of 300 taka per month ($TR = 7 \text{ taka} \times 100 = 700 \text{ taka}$; $TC = 500 \text{ taka} + 5 \text{ taka} \times 100 = 1,000 \text{ taka}$). However, if the firm stops producing, it loses 500 taka per month. Try to draw a graph that represents this relationship. Therefore, even while making a loss, it makes sense for the firm to keep producing.

It should be obvious at this juncture that a firm will only supply goods to the market if the MR is greater or equal to AVC . And since the profit-maximizing condition is $MR=MC$, we now have a new way of deriving a firm's supply curve, as shown in figure 6.8. The horizontal part represents the second scenario ($ATC > AVC > price$) where the firm temporarily shuts-down and there is no supply to the market. In the diagram, that lasts until price rises to 17, which

Figure 6.8: A Firm's Supply Curve



is the lowest point of the average variable cost curve. At price 17, $AVC=P=MR$. The variable cost of production is exactly equal to the price and the firm is indifferent between production. Either scenario gives him the same outcome. Therefore, he may or may not produce. However, once price exceeds 17, we see the familiar upward-sloping supply curve.

But now we have a second way of understanding why the supply curve slopes upward. Look at the first panel of figure 6.8. As MR (price) increases, the firm's profit-maximizing condition ($MR=MC$) changes. As MR goes up, the firm produces more to maximize its profit (recall the marginal analysis). This then is the reason why a supply curve shifts up.⁸

6.3 End of ECO101

There is a lot more left to talk about perfect competitions. For example, we have not talked at all about long-run scenarios. But for this course, we will leave it here. It may seem odd to you that we spent so much time in class discussing this chapter but in terms of number of pages, this is the shortest chapter of this document. That is because I believe that we are at a sufficiently advanced stage in this semester now where I no longer have to explain everything to you. I have merely provided the broad brushstrokes in this and the previous chapter. It is up to you now to do the rest. You have the book. You have Google. You have a department full of professors and lecturers (including, of course, your course instructor) who will be happy to help you out to the best of their abilities. Make use of all the resources that have been made available to you.

/ECO102 notes redacted

⁸If you want to find out why a demand curve shifts downward, you are in for a treat. It involves mapping a person's preference-structure onto a 2-dimension euclidean-space and then introducing a budget-constraint in the analysis. It's super interesting. However, you will need to enroll in ECO206 to learn that.

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